

EXERCISE (O-1)

SINGLE CORRECT TYPE QUESTIONS

Lattice Enthalpy; Condition for Formation of Ionic Bond

- Lattice enthalpy is the change in energy that occurs when of an ionic solid is separated into isolated ions in the gas phase.
(A) One mole (B) One gram (C) Two mole (D) All of these
CCB095
- Which of the following compounds will show the highest lattice energy?
(A) KF (B) NaF (C) CsF (D) RbF
CCB097
- The lattice energy of the lithium compounds is in the following order :
(A) $\text{LiF} > \text{LiCl} > \text{LiBr} > \text{LiI}$ (B) $\text{LiCl} > \text{LiF} > \text{LiBr} > \text{LiI}$
(C) $\text{LiBr} > \text{LiCl} > \text{LiF} > \text{LiI}$ (D) $\text{LiI} > \text{LiBr} > \text{LiCl} > \text{LiF}$
CCB098
- The order of increasing lattice energy of the following compounds, is :
(A) $\text{NaCl} < \text{CaO} < \text{NaBr} < \text{BaO}$ (B) $\text{NaBr} < \text{NaCl} < \text{BaO} < \text{CaO}$
(C) $\text{NaCl} > \text{NaBr} < \text{BaO} < \text{CaO}$ (D) $\text{NaBr} < \text{NaCl} < \text{CaO} < \text{BaO}$
CCB103

Covalent Bond; Types of Electron Pair; Bond Pair and Lone Pair

- Covalent bond is formed by :
(A) Combination of ions (B) Combination of half filled orbitals
(C) Exchange of electrons by atom (D) None of above
CCB116
- The compound which contains ionic as well as covalent bonds is
(A) $\text{C}_2\text{H}_4\text{Cl}_2$ (B) CH_3I (C) KCN (D) H_2O_2
CCB121

Coordinate Bond and Types of Coordinate Bond

- The bonds present in N_2O_5 are :
(A) Only ionic (B) Covalent & coordinate
(C) Only covalent (D) Covalent & ionic
CCB006
- Which of the following is an ionic compound?
(A) CH_4 (B) H_2O (C) CaCl_2 (D) NH_3
CCB113
- The pair of compounds which can form a co-ordinate bond is :
(A) $(\text{C}_2\text{H}_5)_3\text{B}$ and $(\text{CH}_3)_3\text{N}$ (B) HCl and HBr
(C) BF_3 and NH_3 (D) (A) & (C) both
CCB124
- The types of bond present in N_2O_4 are
(A) only covalent (B) only ionic (C) ionic and covalent (D) covalent & coordinate
CCB125

11. Which of the following molecule does not have coordinate bonds?

- (A) $\text{CH}_3\text{-NC}$ (B) CO (C) O_3 (D) CO_3^{2-}

CCB126

Kossel -Lewis approach to Chemical Bonding; Lewis Dot Structure

12. The total number of lone pairs in a chlorine molecule is :

- (A) Six (B) Three (C) Four (D) Seven

CCB118

13. The bond that exists between NH_3 and BF_3 is called

- (A) Electrovalent (B) Covalent
(C) Coordinate (D) Hydrogen

CCB130

14. Which has a coordinate bond (draw LDS)

- (A) SO_3^{2-} (B) CH_4 (C) CO_2 (D) NH_3

CCB131

15. The number of dative bonds in nitric acid molecule is :

- (A) 0 (B) 1 (C) 2 (D) 4

CCB133

16. Indicate the nature of bonding in CCl_4 and CaH_2

- (A) Covalent in CaH_2 and electrovalent in CaH_2
(B) Electrovalent in both CCl_4 and CaH_2
(C) Covalent in both CCl_4 and CaH_2
(D) Electrovalent in CCl_4 and covalent in CaH_2

CCB134

17. Which species does not exists :

- (A) AlF_6^{3-} (B) BF_4^+ (C) BeF_4^{2-} (D) CCl_6^{2-}

CCB137

18. According to LDS, dative bond is present in :

- (A) SO_3 (B) NH_3 (C) K_2CO_3 (D) BF_3

CCB138

Formal Charge

19. The formal charges on the three O-atoms in O_3 molecule are:

- (A) 0, 0, 0 (B) 0, 0, -1 (C) 0, 0, +1 (D) 0, +1, -1

CCB217

Limitations of Lewis Dot Structure (Hypervalent, Hypovalent, Odd Electron Species)

20. The octet rule is not valid for the molecule :

- (A) CO_2 (B) NH_3 (C) SF_2 (D) AlCl_3

CCB142

21. Which of the following pair has electron deficient compounds ?

- (A) BCl_3 , AlCl_3 (B) C_2H_6 , PCl_6^+
(C) SF_2 , Cl_2O (D) BF_4^+ , ICl

CCB143

22. Which is not an exception to octet rule ?
 (A) BF_3 (B) SnCl_4 (C) BeI_2 (D) ClO_2 CCB144
23. Which of the following is an example of hypovalent species?
 (A) BeCl_2 (B) PCl_5 (C) CO_2 (D) SF_6 CCB146
24. Which of the following species are hypervalent?
 (1) PCl_5 , (2) BF_3 , (3) XeF_2 , (4) CO_3^{2-}
 (A) 1, 2, 3 (B) 1, 3 (C) 3, 4 (D) 1, 2 CCB147

Introduction to Valence Bond Theory; Covalency

25. In allene structure ($\text{H}_2\text{C}=\text{C}=\text{CH}_2$) three carbon atoms are joined by :
 (A) Three sigma bonds and three pi bonds
 (B) Two sigma bonds and one pi bond
 (C) Two sigma bonds and two pi bonds
 (D) Three pi bonds only CCB119
26. Number of σ and π bonds present in $\text{CH}_3 - \text{CH} = \text{CH} - \text{C} \equiv \text{CH}$ are :
 (A) $10\sigma, 3\pi$ (B) $10\sigma, 2\pi$ (C) $9\sigma, 2\pi$ (D) $8\sigma, 3\pi$ CCB120
27. Variable covalency is exhibited by :
 (A) P and S (B) N and O (C) N and P (D) F and Cl CCB115
28. In which of the excitation state of Chlorine, ClF_3 is formed?
 (A) In ground state (B) In third excitation state
 (C) In first excitation state (D) In second excitation state CCB158
29. Which compound of xenon is not possible?
 (A) XeF_2 (B) XeF_4 (C) XeF_5 (D) XeF_6 CCB159
30. Nitrogen does not form NF_5 because :
 (A) Nitrogen is member of V group (B) It contains no empty d-orbital
 (C) The bond energy of $\text{N} \equiv \text{N}$ is very high (D) Fluorine is highly electronegative CCB160
31. Which of the following compound is formed in the second excitation state of sulphur atom ?
 (A) SF_4 (B) SF_6 (C) SF_2 (D) None CCB161

Types of Covalent Bond and Overlapping of Orbitals; Orbital Overlap along Different Axes

32. A sigma bond is formed by the overlapping of :
 (A) s-s orbital alone
 (B) s and p orbitals alone
 (C) s-s, s-p or p-p orbitals along internuclear axis
 (D) p-p orbital along the sides CCB163

33. The ratio of number of σ bond to π bond in N_2 and CO molecules are :
 (A) 2.0, 2.0 (B) $2, \frac{1}{2}$ (C) $\frac{1}{2}, \frac{1}{2}$ (D) $\frac{1}{2}, 2$ CCB166
34. No. of σ and π bonds in $C_2(CN)_4$ are respectively :
 (A) $9\sigma, 9\pi$ (B) $8\sigma, 7\pi$ (C) $1\sigma, 1\pi$ (D) $9\sigma, 8\pi$ CCB167
35. If x – axis is the approaching axis between two atoms, then which of the set of orbitals can form π -bond between two atoms in general ?
 (A) $s + px$ (B) $p_y + p_y$ (C) $p_x + p_x$ (D) $p_y + p_z$ CCB168
36. Which of the set of orbitals can form ' δ ' bond between two atoms?
 (A) $d_{yz} + d_{yz}$ along y – axis. (B) $d_{yz} + d_{yz}$ along z – axis.
 (C) $d_{yz} + d_{yz}$ along x – axis. (D) $d_{xz} + d_{xz}$ along x – axis. CCB169
37. Select the **CORRECT** statement if internuclear axis is y-axis :-
 (A) d_{xy} and d_{xy} orbital of two atoms form π -bond
 (B) p_z and p_z orbital of two atoms form σ -bond
 (C) $d_{x^2-y^2}$ and $d_{x^2-y^2}$ orbital of two atoms form π -bond
 (D) p_y and d_{zx} orbital of two atoms form π -bond CCB177
38. The number of sigma and pi bonds in 1-butene-3-yne are
 (A) 5 sigma and 5 pi (B) 7 sigma and 3 pi
 (C) 8 sigma and 2 pi (D) 6 sigma and 4 pi CCB178

Order of Relative Strength of Covalent Bonds

39. Nitrogen forms N_2 but phosphorus do not forms P_2 , but it exists as P_4 . The reason for this is :
 (A) Triple bond is present between phosphorus atoms
 (B) $p\pi - p\pi$ bonding is weak
 (C) $p\pi - p\pi$ bonding is strong
 (D) Multiple bond is formed easily CCB171
40. The correct order of decreasing bond energy is :
 (A) $O-O > S-S > Se-Se$ (B) $C-C > Si-Si > Ge-Ge$
 (C) $F-F > O-O > N-N$ (D) $F-F > Cl-Cl > Br-Br$ CCB172
41. The strength of bonds by $2s - 2s$, $2p - 2p$ and $2p - 2s$ overlapping has the order :
 (A) $2s - 2s > 2p - 2p > 2s - 2p$ (B) $2s - 2s > 2p - 2s > 2p - 2p$
 (C) $2p - 2p > 2s - 2p > 2s - 2s$ (D) $2p - 2p > 2s - 2s > 2p - 2s$ CCB173
42. Which of the following bond is strongest?
 (A) $1s - 1s$ (B) $2p - 2p$ (C) $2s - 2p$ (D) $1s - 2p$ CCB174

43. Consider the following statements :

- I. A sigma (σ) bond is formed when two s-orbitals overlap
 II. A pi (π) bond is formed when two p-orbitals axially overlap
 III. A σ - bond is weaker than π - bond

Which of the above statements is/are correct ?

- (A) I and II (B) II and III (C) I alone (D) II alone

CCB175

Need of Hybridisation; Introduction of Hybridisation

44. Which of the following are arranged in increasing order of length of the hybrid orbitals ?

- (A) $sp < sp^2 < sp^3$ (B) $sp^3 < sp^2 < sp$ (C) $sp^2 < sp^3 < sp$ (D) $sp^2 < sp < sp^3$

CCB183

45. Which of the following is incorrect about hybridization ?

- (A) The concept of hybridization is not applied to isolated atoms.
 (B) Hybridization is the mixing of at least two non-equivalent atomic orbitals.
 (C) The number of hybrid orbitals generated is more than the number of pure atomic orbitals that participate in the hybridization process.
 (D) Number of hybrid orbitals formed is same as that of number of orbitals participate in mixing.

CCB186

46. In the context of carbon, which of the following is arranged in the correct order of electronegativity :

- (A) $sp > sp^2 > sp^3$ (B) $sp^3 > sp^2 > sp$ (C) $sp^2 > sp > sp^3$ (D) $sp^3 < sp < sp^2$

CCB197

47. Select the **CORRECT** set of statements :

As the s-character of a hybrid orbital decreases

- (I) The bond angle decreases (II) The bond strength increases
 (III) The bond length increases (IV) Size of orbital increases
 (A) (I), (III) and (IV) (B) (II), (III) and (IV) (C) (I) and (II) (D) All are CORRECT

CCB247

Prediction of Hybridisation; Types of Hybridisation

48. In following hydrocarbon which one has 75% p – character and 25% s – character :

- (A) C_2H_4 (B) C_2H_2 (C) CH_4 (D) All of the above

CCB185

49. The type of hybrid orbitals used by chlorine atom in $ClO^\ominus$, $ClO_2^\ominus$, $ClO_3^\ominus$ and $ClO_4^\ominus$ is/are :

- (A) sp , sp^2 , sp^3 and sp^3d (B) sp and sp^3
 (C) Only sp^3 (D) Only sp

CCB187

50. In the series, ethane, ethylene and acetylene, the carbon-hydrogen bond length is :

- (A) Equal in all the three (B) Largest in ethane
 (C) Smallest in ethylene (D) Largest in acetylene

CCB188

51. The hybridization of phosphorus in $POCl_3$ is the same as :

- (A) P in PCl_3 (B) S in SF_4
 (C) Cl in ClF_3 (D) B in BCl_3

CCB189

52. The hybridization state of the central atom in HgCl_2 is :
 (A) sp (B) sp^2 (C) sp^3 (D) dsp^2 CCB191
53. Which of the following compounds has four bond pair and one lone pair ?
 (A) NH_4^+ (B) ICl_4^+ (C) SF_4 (D) XeF_4 CCB192
54. Choose the correct option regarding hybridisation of underlined atom :
 (A) $\underline{\text{Xe}}\text{F}_2 \rightarrow sp$ (B) $\underline{\text{Xe}}\text{F}_4 \rightarrow sp^3$ (C) $\underline{\text{P}}\text{OCl}_3 \rightarrow sp^3$ (D) $\underline{\text{I}}\text{F}_5 \rightarrow sp^3d$ CCB198
55. C-H bond distance is the longest in:
 (A) C_2H_2 (B) C_2H_4 (C) C_2H_6 (D) $\text{C}_2\text{H}_2\text{Br}_2$ CCB244

Calculation of Number of p pi-p pi and p pi-d pi Bonds

56. Which of the following has $p\pi - d\pi$ bonding
 (A) NO_3^- (B) CO_3^{2-} (C) BO_3^{3-} (D) SO_3^{2-} CCB179
57. XeO_4 molecule is tetrahedral having :
 (A) Two $p\pi - d\pi$ bonds (B) Four $p\pi - d\pi$ bonds
 (C) One $p\pi - d\pi$ bond (D) Three $p\pi - d\pi$ bonds CCB194

Atomic Orbitals Involved in Hybridisation

58. SeF_6 is sp^3d^2 hybridised and octahedral in shape, which d - orbitals are involved in hybridisation ?
 (A) $d_{x^2-y^2}, d_{xy}$ (B) $d_{x^2-y^2}, d_{z^2}$ (C) d_{xy}, d_{yz} (D) d_{z^2}, d_{xy} CCB195
59. IF_7 is sp^3d^3 hybridised and pentagonal bipyramid in shape, which d - orbitals are involved in hybridisation ?
 (A) d_{xy}, d_{yz}, d_{xz} (B) $d_{x^2-y^2}, d_{z^2}, d_{xy}$
 (C) $d_{x^2-y^2}, d_{yz}, d_{xz}$ (D) $d_{x^2-y^2}, d_{z^2}, d_{yz}$ CCB196

Solid State Hybridisation

60. What is the difference between bond angles in cationic species of PCl_5 and PBr_5 in solid state ?
 (A) 60° (B) $109^\circ 28'$ (C) 0° (D) 90° CCB235
61. What is the hybridisation state of Cl in cationic part of solid Cl_2O_6 ?
 (A) sp^3 (B) sp (C) sp^2 (D) sp^3d CCB251

Introduction of VSEPR Theory

62. Which of the following orders of repulsive interactions is correct ?
 (A) $lp - lp > lp - bp > bp - bp$ (B) $bp - bp > lp - bp > lp - lp$
 (C) $bp - bp > lp - lp > lp - bp$ (D) $lp - lp > bp - bp > lp - bp$ CCB206

VSEPR Table

63. The shape of $\text{IF}_4^\oplus$ will be :
 (A) Square planar (B) Tetrahedral
 (C) Pentagonal bipyramidal (D) See-saw shape
CCB005
64. The Shape of SF_4 is :
 (A) Tetrahedral (B) trigonal bipyramidal
 (C) See – Saw (D) Octahedral
CCB207
65. Choose correct matching:
 (A) $\text{SO}_2 \rightarrow$ Linear (B) $\text{BF}_3 \rightarrow$ trigonal planar
 (C) $\text{PCl}_5 \rightarrow$ octahedral (D) $\text{SF}_6 \rightarrow$ tetrahedral
CCB208
66. The shape of $\text{ClO}_3^\ominus$ ion according to Valence Shell Electron Pair Repulsion (VSEPR) theory will be :
 (A) Triangular planar (B) Pyramidal (C) Tetrahedral (D) Square planar
CCB209
67. The geometry of XeO_3F_2 is :
 (A) Triangular planar (B) Trigonal bipyramidal
 (C) Square planar (D) Tetrahedral
CCB210
68. Which of the set of species have same hybridisation state but different shapes ?
 (A) $\text{NO}_2^\oplus, \text{NO}_3^\ominus, \text{NO}_2^\ominus$ (B) $\text{ClO}_4^\ominus, \text{SF}_4, \text{XeF}_4$
 (C) $\text{NH}_4^\oplus, \text{H}_3\text{O}^\oplus, \text{OF}_2$ (D) $\text{SO}_4^{2\ominus}, \text{PO}_4^{3\ominus}, \text{NH}_4^\oplus$
CCB211
69. Which of the following species are expected to be planar?
 (a) NH_3 (b) $\text{NH}_3^{2\oplus}$ (c) $\text{CH}_3^\oplus$ (d) PCl_3
 The correct answer is :
 (A) b and c (B) c and d (C) b and d (D) a and d
CCB214
70. Amongst $\text{NO}_3^\ominus, \text{AsO}_3^{3\ominus}, \text{CO}_3^{2\ominus}, \text{ClO}_3^\ominus, \text{SO}_3^{2\ominus}$ and $\text{BO}_3^{3\ominus}$, In which set all species are non planar :
 (A) $\text{CO}_3^{2\ominus}, \text{SO}_3^{2\ominus}$ and $\text{BO}_3^{3\ominus}$ (B) $\text{AsO}_3^{3\ominus}, \text{ClO}_3^\ominus$ and $\text{SO}_3^{2\ominus}$
 (C) $\text{NO}_3^\ominus, \text{CO}_3^{2\ominus}$ and $\text{BO}_3^{3\ominus}$ (D) $\text{SO}_3^{2\ominus}, \text{NO}_3^\ominus$ and $\text{BO}_3^{3\ominus}$
CCB216
71. Which of the following has a geometry different from the other three species (having the same geometry)?
 (A) $\text{BF}_4^\ominus$ (B) $\text{SO}_4^{2\ominus}$ (C) XeF_4 (D) $\text{PH}_4^\oplus$
CCB218
72. In the protonation of H_2O , change occurs in :
 (A) Hybridisation state of oxygen (B) Shape of molecule
 (C) Hybridisation and shape both (D) None
CCB220

73. The bond angle at oxygen and hybridization of oxygen in ether (CH_3OCH_3) is :
 (A) $106^\circ 51'$, sp^3 (B) $104^\circ 31'$, sp^3 (C) 110° , sp^3 (D) $109^\circ 28'$, sp^3

CCB245

Maximum Atoms in A Plane; Number of Such Planes

74. Calculate the maximum number of atoms lying in a plane for NH_4^+ :
 (A) 4 (B) 5 (C) 3 (D) 2

CCB215

Bond Parameters; Bond Enthalpy

75. Maximum bond energy is in :
 (A) F_2 (B) N_2 (C) O_2 (D) equal

CCB176

Bond Parameters; Bond length

76. Which of the following has been arranged in order of decreasing bond length ?
 (A) $\text{P-O} > \text{Cl-O} > \text{S-O}$ (B) $\text{P-O} > \text{S-O} > \text{Cl-O}$
 (C) $\text{S-O} > \text{Cl-O} > \text{P-O}$ (D) $\text{Cl-O} > \text{S-O} > \text{P-O}$

CCB246

Bond Parameters; Bond Angle

77. The order of increasing adjacent bond angle in the molecules BeCl_2 , BCl_3 , CCl_4 and SF_6 is :
 (A) $\text{SF}_6 < \text{CCl}_4 < \text{BCl}_3 < \text{BeCl}_2$ (B) $\text{BeCl}_2 < \text{BCl}_3 < \text{CCl}_4 < \text{SF}_6$
 (C) $\text{SF}_6 < \text{CCl}_4 < \text{BeCl}_2 < \text{BCl}_3$ (D) $\text{BCl}_3 < \text{BeCl}_2 < \text{SF}_6 < \text{CCl}_4$

CCB190

78. The bond angle in H_2O molecule is less than that of NH_3 molecule because :
 (A) The hybridisation of O in H_2O and N in NH_3 is different
 (B) The atomic radii of N and O are different
 (C) There is one lone pair of electrons on O and two lone pairs of electrons on N
 (D) There are two lone pairs of electrons on O and one lone pair of electrons on N

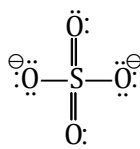
CCB212

Concept of Resonance; Bond Order; Comparison of Bond Length and Bond Strength using Bond Order

79. The weakest Cl -O bond found in:
 (A) $\text{ClO}_4^\ominus$ (B) $\text{ClO}_2^\ominus$
 (C) $\text{ClO}^\ominus$ (D) $\text{ClO}_3^\ominus$

CCB149

80. One of the resonating structure of SO_4^{2-} is :

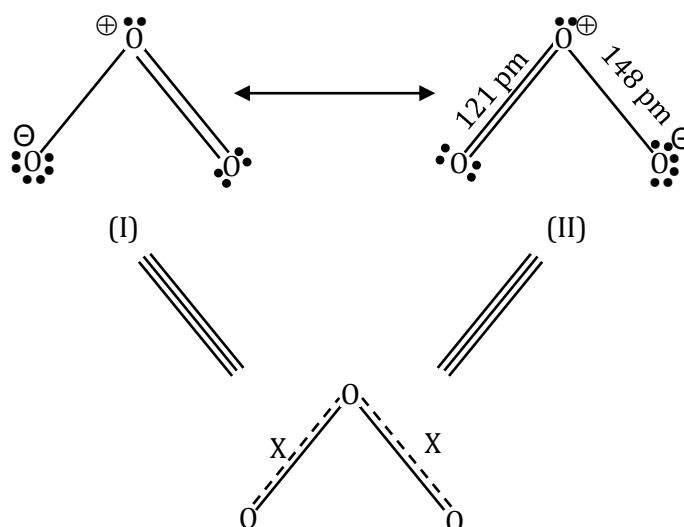


Which set of average formal charge on oxygen and bond order is correct ?

- (A) -0.5 and 1.5 (B) -1.5 and 3
 (C) -2 and 3 (D) -1.5 and 1.5

CCB150

81. O_3 (ozone) molecule can be equally represented by the structures I and II shown below :



Which of the following value of x is correct ?

- (A) 148 pm (B) 121 pm
(C) between 121 pm and 148 pm (D) more than 148 pm

CCB152

82. Point out incorrect statement about resonance

- (A) Resonance structures should have equal energy
(B) In resonance structures, the constituent atoms should be in the same position
(C) In resonance structures, there should not be the same number of electron pairs
(D) Resonance structures should differ only in the location of electrons around the constituent atoms

CCB154

Bent's Rule

83. Choose the INCORRECT statements :

- (A) $d_{P-F}(\text{axial}) > d_{P-F}(\text{equatorial})$ in PF_3Cl_2
(B) $d_{P-F} > d_{P-Cl}$ in PF_2Cl_3
(C) $d_{P-Cl}(PF_2Cl_3) > d_{P-Cl}(PF_3Cl_2)$
(D) All d_{P-Cl} (in PF_2Cl_3) are identical while all d_{P-F} (in PF_3Cl_2) are not identical

CCB236

84. Order of C – H bond lengths :

- (A) $d_{C-H}(CH_4)$ (B) $d_{C-H}(CH_3F)$
(C) $d_{C-H}(CH_2F_2)$ (D) $d_{C-H}(CHF_3)$
(A) $A > B > C > D$ (B) $D > C > B > A$
(C) $A > D > B > C$ (D) $D > A > B > C$

CCB237

85. In which of the following molecule C—C bond length will be highest?

- (A) $CF_3 - CF_3$ (B) $F_2CH - CHF_2$
(C) $FCH_2 - CH_2F$ (D) $CH_3 - CF_3$

CCB248

Drago's Rule

86. Among the following, the CORRECT statement is :
 (A) Between NH_3 and PH_3 , NH_3 is a better electron donor because the lone pair of electrons occupies spherical 's' orbital and is less directional
 (B) Between NH_3 and PH_3 , PH_3 is a better electron donor because the lone pair of electrons occupies sp^3 orbital and is more directional
 (C) Between NH_3 and PH_3 , NH_3 is a better electron donor because the lone pair of electrons occupies sp^3 orbital and is more directional
 (D) Between NH_3 and PH_3 , PH_3 is a better electron donor because the lone pair of electrons occupies spherical 's' orbital and is less directional

CCB240

87. Which of the following order is **CORRECT** for increasing p-character in orbital used for bonding by central atom
 (A) $\text{SiH}_4 > \text{CH}_4$ (B) $\text{H}_2\text{S} > \text{H}_2\text{O}$ (C) $\text{PH}_4^+ > \text{PH}_3$ (D) $\text{NH}_3 > \text{PH}_3$

CCB250

88. The species in which the central atom uses sp^2 hybrid orbitals in its bonding is
 (A) PH_3 (B) NH_3 (C) H_3C^+ (D) SbH_3

CCB252

Odd Electron Molecules

89. An oxide of chlorine which is an odd electron molecule is:
 (A) ClO_3 (B) Cl_2O_6 (C) Cl_2O_7 (D) Cl_2O
 90. In which of the following processes, the magnetic behaviour of the species is changed ?
 (A) $2\text{CH}_3 \rightarrow \text{C}_2\text{H}_6$ (B) $2\text{NO}_2 \rightarrow \text{N}_2\text{O}_4$
 (C) $2\text{ClO}_3 \rightarrow \text{Cl}_2\text{O}_6$ (D) All of these

CCB145

CCB241

91. Which of the following odd e^- species is planar ?
 (A) $\dot{\text{C}}\text{H}_3$ (B) $\dot{\text{C}}\text{IO}_3$ (C) $\dot{\text{C}}\text{F}_3$ (D) None of these

CCB242

Molecules which Do Not Exist

92. Which of the following molecules do not exist ?
 (A) BF_6^{2+} (B) NCl_3 (C) NOF_3 (D) XeF_5^+
 93. Select non existing species :
 (A) PH_3 (B) PH_4^+ (C) $[\text{PF}_6]^-$ (D) None of these
 94. Select non existing species
 (A) PF_5 (B) SF_6 (C) PH_5 (D) SF_4
 95. Among the fluorides below, the one which does not exist is -
 (A) XeF_4 (B) BeF_4 (C) SF_4 (D) CF_4

CCB259

CCB260

CCB261

CCB262

96. Which of the following compound exist -

- (A) XeH_2 (B) ClF_7 (C) PH_5 (D) ICl_3

CCB263

Dipole Moment; Calculation of Percent Ionic Character

97. The experimental value of the dipole moment of HCl is 1.03 D. The length of the H – Cl bond is 1.275 Å. The percentage of ionic character in HCl is nearly :

- (A) 43 (B) 21 (C) 17 (D) 7

CCB269

Polarity of Species

98. The dipole moment of NH_3 is :

- (A) Less than dipole moment of NCl_3 (B) Higher than dipole moment of NCl_3
(C) Equal to the dipole moment of NCl_3 (D) None of these

CCB264

99. Species having zero dipole moment :

- (A) XeF_4 (B) 1,2,4 trichloro benzene
(C) SF_4 (D) CH_2Cl_2

CCB266

100. The correct order of dipole moment is :

- (A) $\text{CH}_4 < \text{NF}_3 < \text{NH}_3 < \text{H}_2\text{O}$ (B) $\text{NF}_3 < \text{CH}_4 < \text{NH}_3 < \text{H}_2\text{O}$
(C) $\text{NH}_3 < \text{NF}_3 < \text{CH}_4 < \text{H}_2\text{O}$ (D) $\text{H}_2\text{O} < \text{NH}_3 < \text{NF}_3 < \text{CH}_4$

CCB267

101. Which of the following has the highest value of dipole moment?

- (A) HCl (B) HF (C) HI (D) HBr

CCB268

102. Which of the following species are polar ?

- (a) C_6H_6 (b) XeF_2 (c) SO_2 (d) SF_4 (e) SF_6

Correct answer is :

- (A) (b) and (d) (B) (a), (b) and (e)
(C) (a) and (e) (D) (c) and (d)

CCB270

103. Which contains both polar and non-polar bonds ?

- (A) NH_4Cl (B) HCN (C) H_2O_2 (D) CH_4

CCB273

104. Find out the incorrect order of the dipole moment among the following pair of compound :

- (A) $\text{NH}_3 > \text{NF}_3$
(B) p-dichloro benzene > o-dichloro benzene
(C) $\text{CH}_3\text{Cl} > \text{CH}_2\text{Cl}_2$
(D) $\text{SiF}_4 < \text{SF}_4$

CCB274

105. In which of the following both are nonpolar?

- (A) XeF_4 , O_3 (B) $\text{XeF}_5^\ominus$, N_2O_4
(C) SF_4 , XeF_2 (D) BrF_3 , SCl_2

CCB275

106. Which of the following molecules is nonpolar ?

- (i) PbCl_4 (ii) BF_3 (iii) SnCl_2 (iv) CS_2
 (A) (i), (ii), (iii) (B) (i), (ii), (iii), (iv) (C) (i), (ii), (iv) (D) (ii), (iii), (iv)

CCB276

107. Which of the following has the highest dipole moment ?

- (A) *o*-Dichlorobenzene (B) *m*-Dichlorobenzene
 (C) *p*-Dichlorobenzene (D) All have equal values

CCB277

108. Both CO_2 and H_2O contain polar covalent bonds but CO_2 is nonpolar while H_2O is polar because :

- (A) H atom is smaller than C atom
 (B) CO_2 is a linear molecule while H_2O is an angular molecule
 (C) O – H bond is more polar than C – H bond
 (D) CO_2 contains multiple bonds while H_2O has only single bonds

CCB278

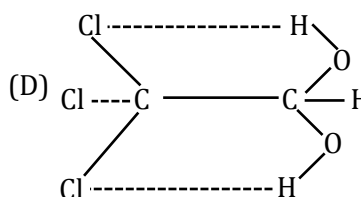
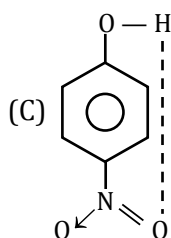
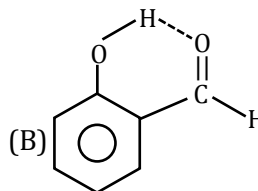
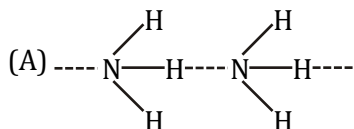
109. BF_3 and NF_3 both molecules are covalent, but BF_3 is non-polar and NF_3 is polar. Its reason is

- (A) In uncombined state boron is metal and nitrogen is gas
 (B) B–F bond has no dipole moment whereas N–F bond has dipole moment
 (C) The size of boron atom is smaller than nitrogen
 (D) BF_3 is planar whereas NF_3 is pyramidal

CCB280

Hydrogen Bonding; Strength of Hydrogen Bonding

110. In which of the following molecule e, the shown hydrogen bond is not possible ?



CCB295

111. Maximum number of H-Bond is shown by :

- (A) H_2O (B) H_2Se (C) H_2S (D) HF

CCB296

112. KF combines with HF to form KHF_2 . The compound contains the species :

- (A) K^+ , F^- and H^+ (B) K^+ , F^- and HF
 (C) K^+ and $[\text{HF}_2]^-$ (D) $[\text{KHF}]^+$ and F_2

CCB298

113. For H_2O_2 , H_2S , H_2O and HF , the correct order of decreasing strength of hydrogen bonding is :

- (A) $\text{H}_2\text{O} > \text{HF} > \text{H}_2\text{O}_2 > \text{H}_2\text{S}$ (B) $\text{HF} > \text{H}_2\text{O}_2 > \text{H}_2\text{O} > \text{H}_2\text{S}$
 (C) $\text{HF} > \text{H}_2\text{O} > \text{H}_2\text{O}_2 > \text{H}_2\text{S}$ (D) $\text{H}_2\text{O}_2 > \text{H}_2\text{O} > \text{HF} > \text{H}_2\text{S}$

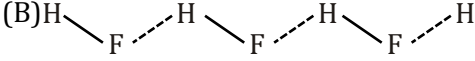
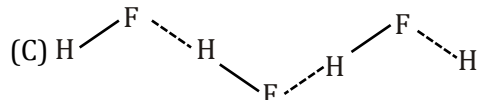
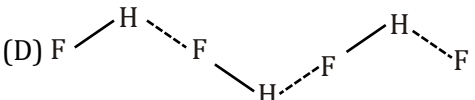
CCB307

114. The order of strength of hydrogen bonds is:

- (A) $\text{NH}\cdots\text{N} > \text{OH}\cdots\text{O} > \text{FH}\cdots\text{F}$ (B) $\text{NH}\cdots\text{N} < \text{OH}\cdots\text{O} < \text{FH}\cdots\text{F}$
 (C) $\text{NH}\cdots\text{N} > \text{OH}\cdots\text{O} < \text{FH}\cdots\text{F}$ (D) $\text{NH}\cdots\text{N} < \text{OH}\cdots\text{O} > \text{FH}\cdots\text{F}$

CCB309

115. The H bond in solid HF can be best represented as:

- (A) $\text{H} - \text{F} \cdots \text{H} - \text{F} \cdots \text{H} - \text{F}$ (B) 
 (C)  (D) 

CCB310

116. In which of the following compounds does hydrogen bonding occur

- (A) SiH_4 (B) LiH (C) HI (D) NH_3

CCB312

Types of Hydrogen Bonding

117. The crystal lattice of ice is mostly formed by :

- (A) Ionic forces (B) Covalent bonds
 (C) Inter molecular H-bonds (D) Covalent as well as H-bonds

CCB297

118. Which of the following compounds would have significant intermolecular hydrogen bonding?

HF , CH_3OH , N_2O_4 , CH_4

- (A) HF , N_2O_4 (B) HF , CH_4 , CH_3OH (C) HF , CH_3OH (D) CH_3OH , CH_4

CCB306

119. Which one of the following does not have intermolecular H-bonding?

- (A) H_2O (B) *o*-nitro phenol
 (C) HF (D) CH_3COOH

CCB308

Applications of Hydrogen Bonding; Viscosity ; Boiling Point; Volatility; Solubility in Water

120. The correct order of volatility is :

- (A) $\text{NH}_3 < \text{H}_2\text{O}$ (B) *p* - nitro phenol < *o* - nitro phenol
 (C) $\text{CH}_3\text{OH} > \text{CH}_3 - \text{O} - \text{CH}_3$ (D) $\text{HF} > \text{HCl}$

CCB299

121. Which of the following hydrides has the lowest boiling point ?

- (A) H_2O (B) H_2S (C) H_2Se (D) H_2Te

CCB300

122. Ethanol has a higher boiling point than dimethyl ether though they have the same molecular weight.

This is due to :

- (A) resonance (B) coordinate bonding
 (C) hydrogen bonding (D) ionic bonding

CCB305

123. In the following which has highest boiling point
 (A) HI (B) HF (C) HBr (D) HCl
 CCB311

124. Acetic acid exists as dimer in benzene due to
 (A) Condensation reaction (B) Hydrogen bonding
 (C) Presence of carboxyl group (D) Presence of hydrogen atom at α - carbon
 CCB313

125. The high density of water compared to ice is due to
 (A) Hydrogen bonding interactions
 (B) Dipole-dipole interactions
 (C) Dipole-induced dipole interactions
 (D) Instantaneous dipole-induced dipole interactions
 CCB314

Inert Pair Effect

126. E represents an element belonging to carbon family.
 $E + 2X_2 \longrightarrow EX_4$ (X = F, Cl, Br, I)
 (A) Stability of EX_4 decreases down the 14th group
 (B) PbI_4 does not exist
 (C) Ge & Pb forms EX_2 as well.
 (D) All are correct
 CCB521

Back Bonding

127. Which of the following statements is not correct ?
 (A) $(CH_3)_3COH$ is less acidic than $(CH_3)_3SiOH$.
 (B) Like CO_2 its analogue of Si is not stable.
 (C) In phosgene, C–O bond length is longer than expected while C–Cl bond length is shorter.
 (D) None of the above.
 CCB485

128. In $B_3N_3H_6$ compound, the correct option is (State 'T' for True and 'F' for False) :
 (P) '3' $p\pi-p\pi$ bond is present.
 (Q) It can show back bonding.
 (R) It is non-polar compound with non-polar bonds.
 (S) More e⁻ density is present at Boron atom.
 (A) TTFF (B) TTTT (C) TTTF (D) FTTF
 CCB486

129. In which of the following cases, the increase in bond angle is not due to back bonding ?
 (A) $\hat{C}OC$ angle in CH_3OCH_3 (B) $\hat{Si}NSi$ angle in $N(SiH_3)_3$
 (C) $\hat{F}BF$ angle in BF_3 (D) $F-\hat{P}-F$ angle in PF_3
 CCB487

130. Which of the following molecule has large extent of back bonding ?
 (A) $O(SiH_3)_2$ (B) OCl_2
 (C) Both have same extent of back bonding (D) None of the above
 CCB488

131. Choose the incorrect statement from the following.
 (A) Bond angles are not affected in BF_3 due to back bonding.
 (B) Bond angles are affected in PF_3 due to back bonding.
 (C) Bond angles are not affected in B(OMe)_3 due to back bonding.
 (D) None of the above

CCB489

Bridge Bonding($3\text{C}-2\text{e}^-$, $3\text{C}-4\text{e}^-$), B_2H_6 , Al_2Cl_6 , Bond Behaviour of B_2H_6 ; Bridge Bonding in other Compounds; Reactions of Diborane

132. Which of the following statement is INCORRECT ?
 (A) Hybridization state of 'Be' in BeH_2 (solid) is sp^3 .
 (B) BeH_2 (solid) is a planar molecule
 (C) BeH_2 (solid) is a electron deficient molecule.
 (D) In BeH_2 (solid) $3\text{c}-2\text{e}^-$ bond present

CCB522

133. Unsymmetrical cleavage of B_2H_6 occurs on reaction with :
 (A) $\text{Me}_3\text{N:}$ (B) $\text{Me}_3\text{P:}$ (C) $\text{Et}_2\ddot{\text{S}}:$ (D) $\text{H}_3\text{N:}$

CCB491

134. The state of hybridization of central atom in dimer form of both BH_3 and BeH_2 is :
 (A) sp^2 , sp (B) sp^3 , sp^2 (C) sp^3 , sp^3 (D) sp^2 , sp^3

CCB492

135. Which of the following molecule has complete octet ?
 (A) B_2H_6 (B) Al_2Cl_6 (C) Be_2Cl_4 (D) BeH_2

CCB493

Oxyacids of Phosphorus

136. Which of the following oxyacid contains P – P bond?
 (A) $\text{H}_4\text{P}_2\text{O}_8$ (B) $\text{H}_4\text{P}_2\text{O}_5$ (C) $\text{H}_4\text{P}_2\text{O}_7$ (D) $\text{H}_4\text{P}_2\text{O}_6$

CCB523

137. How many of the following molecules have maximum number of P–O–P linkage?
 (A) $\text{H}_5\text{P}_3\text{O}_{10}$ (B) $\text{H}_5\text{P}_5\text{O}_{15}$ (C) $\text{H}_7\text{P}_5\text{O}_{16}$ (D) $\text{H}_4\text{P}_2\text{O}_7$

CCB495

138. The molecular formula of sodium hydrogen phosphate is :
 (A) NaH_2PO_4 (B) Na_2HPO_4 (C) NaH_2PO_3 (D) Na_2HPO_3

CCB496

Oxyacids of Sulphur

139. The compound having S—S linkage is :
 (A) $\text{H}_2\text{S}_2\text{O}_7$ (B) $\text{H}_2\text{S}_2\text{O}_8$ (C) $\text{H}_2\text{S}_2\text{O}_6$ (D) $(\text{SO}_3)_3$

CCB497

Oxyacids of Chlorine

140. $\text{Cl}=\text{O}$ bond is NOT present in:
 (A) Chloric acid (B) Hypochlorous acid (C) Chlorous acid (D) Perchloric acid
141. Which of the following compound does not have peroxy linkage?
 (A) Marshall's acid (B) Caro's acid (C) Perchloric acid (D) Pernitric acid

CCB524

CCB498

Hydrolysis

142. CCl_4 is inert towards hydrolysis but SiCl_4 is readily hydrolysed because
 (A) carbon cannot expand its octet but silicon can expand its octet
 (B) ionisation potential of carbon is higher than silicon
 (C) carbon forms double and triple bonds
 (D) electronegativity of carbon is higher than that of silicon
 CCB525
143. Which one of the following compounds gives methane on treatment with water :-
 (A) Al_4C_3 (B) CaC_2 (C) Mg_2C_3 (D) SiC
 CCB526
144. Find the product(s) of hydrolysis of pyrophosphorous acid $\left(\text{HO}-\overset{\text{O}}{\underset{\text{H}}{\text{P}}}-\text{O}-\overset{\text{O}}{\underset{\text{H}}{\text{P}}}-\text{OH} \right)$.
 (A) $2\text{H}_3\text{PO}_4$ (B) $\text{H}_3\text{PO}_4 + \text{H}_3\text{PO}_3$ (C) $2\text{H}_3\text{PO}_3$ (D) $\text{H}_3\text{PO}_3 + \text{H}_3\text{PO}_2$
 CCB499
145. Type of hybridization of transition state of SeF_4 , when it undergoes hydrolysis is :
 (A) sp^3d (B) sp^3d^2 (C) dsp^3 (D) sp^3
 CCB500
146. SbCl_3 and BiCl_3 on hydrolysis gives :
 (A) Sb^{+3} and Bi^{+3} (B) $\text{Sb}(\text{OH})_3$ and $\text{Bi}(\text{OH})_3$
 (C) SbOCl and BiOCl (D) None of these
 CCB501
147. Calculate the moles of oxyacid formed when 18 gram of water reacts with NCl_3 and undergoes complete hydrolysis.
 (A) 1 mole (B) 3 mole
 (C) $\frac{1}{3}$ mole (D) $\frac{1}{2}$ mole
 CCB502
148. Which of the following statements is correct ?
 (A) On hydrolysis, SeF_6 and TeF_6 four water molecules are consumed.
 (B) On hydrolysis of PCl_3 , product obtained has basicity 3.
 (C) On hydrolysis of SF_4 , hybridisation of S in transition state is sp^3d .
 (D) All molecules of BF_3 are not hydrolysed in water.
 CCB503
149. What is the oxidation number of P in product when POCl_3 is completely hydrolysed?
 (A) +5 (B) +3 (C) +4 (D) +2
 CCB504

Introduction of Weak Forces; Types of Weak Forces; Factors Affecting London Dispersion Forces; Comparison of Boiling Point of Species

150. Which is the weakest among the following types of bonds?
 (A) Debye force (B) Metallic bond
 (C) Dipole-dipole interaction (D) Hydrogen bond
 CCB319

151. Which of the following inert gas liquefied easily as compare to others ?
(A) Kr (B) He (C) Ne (D) Ar CCB322
152. The relative strength of interionic/intermolecular forces in decreasing order is :
(A) Ion-dipole > Ion-ion > Dipole-dipole (B) Dipole-dipole > Ion-dipole > Ion-ion
(C) Ion-dipole > Dipole-dipole > Ion-ion (D) Ion-ion > Ion-dipole > Dipole-dipole CCB323
153. Which of the following boiling point order is correct?
(A) He > T₂ > D₂ (B) He < T₂ < D₂ (C) T₂ > He > D₂ (D) He < D₂ < T₂ CCB324
154. Identify the incorrect order of boiling point in the following pairs :
(A) B(OH)₃ < B(OCH₃)₃ (B) NF₃ < N(CH₃)₃
(C) BF₃ < B(CH₃)₃ (D) C₂H₆ < C₂F₆ CCB325
155. Select the CORRECT statement :
(A) Boiling point of NF₃ is greater than NMe₃
(B) Greater the dipole moment in molecule, greater will be the dipole-dipole interaction between the molecules.
(C) London dispersion force increases with decreasing number of electrons
(D) Boiling point of hydrides of carbon family decreases down the group. CCB327
156. In solid argon, the atoms are held together by
(A) Ionic bonds (B) Hydrogen bonds
(C) Van Der Waals forces (D) Hydrophobic forces CCB329
157. Which of the following exhibits the weakest intermolecular forces
(A) He (B) HCl (C) NH₃ (D) H₂O CCB331
158. The intermolecular interaction that is dependent on the inverse cube of distance between the molecules is :
(A) London force (B) Hydrogen bond
(C) Ion-ion interaction (D) Ion-dipole interaction CCB333
159. Which one has the highest boiling point ?
(A) Kr (B) Xe
(C) He (D) Ne CCB334

Molecular Orbital Theory; Drawbacks of Valence Bond Theory; Introduction of Molecular Orbital Theory

160. During the formation of a molecular orbital from atomic orbitals, probability of electron density is
(A) minimum in the nodal plane (B) maximum in the nodal plane
(C) zero in the nodal plane (D) zero on the surface of the lobe CCB351

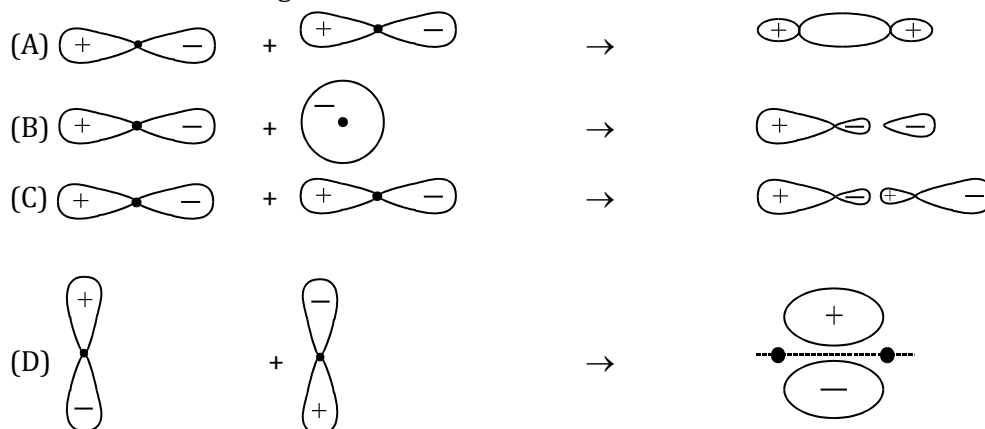
161. Out of the following which has smallest bond length

- (A) O_2 (B) O_2^+ (C) O_2^- (D) O_2^{2-}

CCB364

Linear Combination of Atomic Orbitals

162. Which of the following combination of orbitals is CORRECT ?



CCB344

163. If the z-axis is taken as the internuclear axis, then which of the following combinations of atomic orbitals is a nonbonding combination?

- (A) s and p_y (B) p_x and s
(C) p_x and p_y (D) All of these

CCB345

164. Which of the following overlapping indicates formation of bonding molecular orbitals having gerade symmetry?



CCB352

165. Which of the following molecular orbitals has two different nodal planes

- (A) σ_{2s} (B) π_{2p_y} (C) $\pi_{2p_y}^*$ (D) $\sigma_{2p_x}^*$

CCB358

Bonding in some homonuclear diatomic molecules (Molecular Orbital Diagram of H_2 , He_2 , Li_2 , O_2 , B_2 , C_2 , N_2 , F_2 , Ne_2)

166. Which of the following does not exist ?

- (A) B_2 (B) C_2 (C) He_2 (D) N_2

CCB343

167. In the formation of N_2^+ , the electron is removed from :

- (A) σ -orbital (B) π -orbital (C) σ^* -orbital (D) π^* -orbital

CCB350

168. Which one of the following species is diamagnetic in nature

- (A) He_2^+ (B) H_2 (C) H_2^+ (D) H_2^0

CCB357

169. LUMO in N_2 is :

- (A) π_{2px}^* (B) σ_{2pz}^* (C) π_{2px} (D) σ_{2pz}

CCB372

170. HOMO in O_2 is :

- (A) π_{2px}^* (B) σ_{2pz}^* (C) π_{2px} (D) σ_{2pz}

CCB373

171. Most stable species is :

- (A) $O_2^{1\oplus}$ (B) $O_2^{1\ominus}$
(C) $O_2^{2\ominus}$ (D) All equal

CCB374

172. The bond energy order of $He_2^{\oplus}$ and $HeH^{\oplus}$ is

- (A) $He_2^{\oplus} > HeH^{\oplus}$ (B) $HeH^{\oplus} = He_2^{\oplus}$
(C) $HeH^{\oplus} > He_2^{\oplus}$ (D) Can't be predicted

CCB376

173. Number of anti bonding electrons in N_2 is :

- (A) 4 (B) 10 (C) 12 (D) 14

CCB378

Applications of MOT(Molecular Behaviour) ; 1 Bond Order; 2 Bond Length; 3 Bond Strength; 4 Ionisation Energy; 5 Colour

174. Which of the following species will have the minimum bond energy :

- (A) N_2 (B) $N_2^{\ominus}$ (C) $N_2^{\oplus}$ (D) $N_2^{2\ominus}$

CCB009

175. Which of the following species will have the minimum bond energy ?

- (A) N_2 (B) $N_2^{\ominus}$ (C) $N_2^{\oplus}$ (D) $N_2^{2\ominus}$

CCB368

176. When N_2 is ionised to $N_2^{\oplus}$ bond length & if O_2 is ionised to $O_2^{\oplus}$ bond length

Select the CORRECT option to fill the blank space respectively:

- (A) Increases and decreases (B) Decreases and increases
(C) Increases and increases (D) Decreases and decreases

CCB369

177. Which of the following order of ionisation energy is correct?

- (A) $F_2 > F$ (B) $F_2 < F$ (C) $F_2 = F$ (D) Cannot predict

CCB370

178. Choose Correct order of ionisation energy ?

- (A) $O_2 > O$ (B) $O_2 < O$ (C) $O_2 = O$ (D) Cannot predict

CCB371

179. The highest bond length is present in :

- (A) B_2 (B) $B_2^{2\ominus}$ (C) $B_2^{1\oplus}$ (D) All equal

CCB375

Concept of Hydration; Hydration Energy; Hydrated Radius; Ionic Mobility in Water; Conductivity in Water; Condition for Solubility of Ionic Compounds in Water

- 180.** Hydrated radius of $M^{\oplus}$ ions in decreasing order :
 (A) $Li^{\oplus}(aq) > Na^{\oplus}(aq) > K^{\oplus}(aq) > Rb^{\oplus}(aq) > Cs^{\oplus}(aq)$
 (B) $Li^{\oplus}(aq) > Na^{\oplus}(aq) < K^{\oplus}(aq) < Rb^{\oplus}(aq) < Cs^{\oplus}(aq)$
 (C) $Li^{\oplus}(aq) > Na^{\oplus}(aq) > K^{\oplus}(aq) > Rb^{\oplus}(aq) < Cs^{\oplus}(aq)$
 (D) $Li^{\oplus}(aq) < Na^{\oplus}(aq) < K^{\oplus}(aq) < Rb^{\oplus}(aq) < Cs^{\oplus}(aq)$
 CCB002
- 181.** The hydration energy of $Mg^{2\oplus}$ is greater than the hydration energy of :
 (A) $Al^{3\oplus}$ (B) $Mg^{3\oplus}$
 (C) $Na^{\oplus}$ (D) $Be^{2\oplus}$
 CCB388
- 182.** Which of the compound is least soluble in water ?
 (A) AgF (B) AgCl
 (C) AgBr (D) AgI
 CCB389
- 183.** Ionic conductance of hydrated $M^{\oplus}$ ions are in the order :
 (A) $Li^{\oplus}(aq) > Na^{\oplus}(aq) > K^{\oplus}(aq) > Rb^{\oplus}(aq) > Cs^{\oplus}(aq)$
 (B) $Li^{\oplus}(aq) > Na^{\oplus}(aq) < K^{\oplus}(aq) < Rb^{\oplus}(aq) < Cs^{\oplus}(aq)$
 (C) $Li^{\oplus}(aq) > Na^{\oplus}(aq) > K^{\oplus}(aq) > Rb^{\oplus}(aq) < Cs^{\oplus}(aq)$
 (D) $Li^{\oplus}(aq) < Na^{\oplus}(aq) < K^{\oplus}(aq) < Rb^{\oplus}(aq) < Cs^{\oplus}(aq)$
 CCB390
- 184.** The ionic mobility of alkali metal ions in aqueous solution is maximum for
 (A) $Li^{\oplus}$ (B) $Na^{\oplus}$ (C) $K^{\oplus}$ (D) $Rb^{\oplus}$
 CCB391
- 185.** Sodium sulphate is soluble in water whereas barium sulphate is sparingly soluble because :
 (a) The hydration energy of Na_2SO_4 is less than its lattice energy
 (b) The hydration energy of Na_2SO_4 is more than its lattice energy
 (c) The lattice energy of $BaSO_4$ is more than its hydration energy
 (d) The lattice energy has no role to play in solubility
 (A) a & b (B) b & c
 (C) c & d (D) b & d
 CCB392
- 186.** Choose the compound from each of the following pairs that has the more solubility in water :-
 (a) $BeSO_4, BaSO_4$ (b) $LiOH, CsOH$ (c) $AgCl, AgI$
 (A) $BeSO_4, CsOH, AgI$ (B) $BeSO_4, CsOH, AgCl$
 (C) $BaSO_4, LiOH, AgCl$ (D) $BaSO_4, LiOH, AgI$
 CCB393
- 187.** Which of the following metal ion will have maximum hydrated radius :
 (A) $Be^{2\oplus}$ (B) $Mg^{2\oplus}$
 (C) $Ca^{2\oplus}$ (D) $Sr^{2\oplus}$
 CCB396

Fajan's Rule; Polarisation; Polarising Power; Polarizability; Factors Affecting Polarisation

188. The charge/size ratio of a cation determines its polarizing power. Which one of the following sequences represents the increasing order of the polarizing power of the cationic species, K^+ , Ca^{2+} , Mg^{2+} , Be^{2+} ?

- (A) $Be^{2+} < K^+ < Ca^{2+} < Mg^{2+}$ (B) $K^+ < Ca^{2+} < Mg^{2+} < Be^{2+}$
 (C) $Ca^{2+} < Mg^{2+} < Be^{2+} < K^+$ (D) $Mg^{2+} < Be^{2+} < K^+ < Ca^{2+}$

CCB399

189. CCl_4 is more covalent than $LiCl$ because :

- (A) There is more polarization of Cl in CCl_4 (B) There is more polarization of Cl in $LiCl$
 (C) CCl_4 has more weight (D) None of above

CCB400

190. Polarization is the distortion of the shape of an anion by an adjacently placed cation. Which of the following statements is correct

- (A) Maximum polarization is brought about by a cation of high charge
 (B) Minimum polarization is brought about by a cation of low radius
 (C) A large cation is likely to bring about a large degree of polarization
 (D) A small anion is likely to undergo a large degree of polarization

CCB408

191. In which of the following molecule is the covalent bond most polar

- (A) HI (B) HBr (C) HCl (D) H_2

CCB409

192. Which of the following has a high polarising power

- (A) Mg^{2+} (B) Al^{3+} (C) Na^+ (D) Ca^{2+}

CCB411

193. According to Fajan's rule, covalent bond is favoured by

- (A) Large cation and small anion (B) Large cation and large anion
 (C) Small cation and large anion (D) Small cation and small anion

CCB413

Properties Affected by Polarisation; Covalent Character of Ionic Compounds

194. Amongst $LiCl$, $RbCl$, $BeCl_2$ and $MgCl_2$ the compounds with the greatest and the least ionic character, respectively are :

- (A) $LiCl$ and $RbCl$ (B) $RbCl$ and $BeCl_2$
 (C) $RbCl$ and $MgCl_2$ (D) $MgCl_2$ and $BeCl_2$

CCB397

195. Compound with maximum ionic character is formed from :

- (A) Na and Cl (B) Cs and F
 (C) Cs and I (D) Na and F

CCB398

196. Select correct order out of given options :

- (A) $BeCO_3 < BaCO_3$: Covalent character (B) $BeO > SrO$: lattice energy
 (C) $Be^{2+} < Li^+$: Hydration energy (D) $Be^{2+} (aq.) > Li^+ (aq.)$: Ionic mobility

CCB402

197. Maximum covalent character is associated with the compound

- (A) NaI (B) MgI₂ (C) AlCl₃ (D) AlI₃

CCB412

Properties Affected by Polarisation; Melting Point of Ionic Compounds

198. Which of the following has highest melting point

- (A) BeCl₂ (B) MgCl₂ (C) CaCl₂ (D) BaCl₂

CCB405

199. Which compound is highest covalent

- (A) LiCl (B) LiF (C) LiBr (D) LiI

CCB407

Properties Affected by Polarisation; Solubility of Ionic Compounds in Water

200. Which of the following set of characteristics lead to the increase in solubility of ionic substances?

- (A) High dipole moment, strong attraction by an ion towards solvent and large solvation energy
 (B) Low dipole moment, weak attraction by an ion towards solvent and high solvation energy
 (C) High dipole moment, strong attraction by an ion towards solvent and low solvation energy
 (D) High dipole moment, weak attraction by an ion towards solvent and large solvation energy

CCB403

EXERCISE (O-2)

MULTIPLE CORRECT TYPE QUESTIONS

Kossel -Lewis approach to Chemical Bonding; Lewis Dot Structure

1. Which of the following compounds contain ionic, covalent and co-ordinate bonds?
 (A) NH_4Cl (B) KCN (C) NaBF_4 (D) NaOH

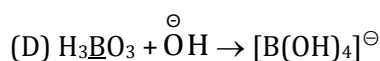
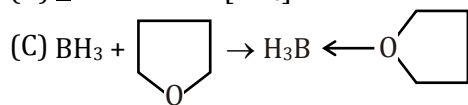
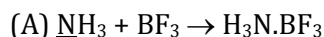
CCB135

Prediction of Hybridisation; Types of Hybridisation

2. Which of the following statement(s) is / are not correct?
 (A) Hybridization is the mixing of atomic orbitals of large energy difference.
 (B) sp^2 -hybrid orbitals are formed from two p - atomic orbitals and one s - atomic orbital
 (C) sp^3d^2 -hybrid orbitals are all at 90° to one another
 (D) sp^3 -hybrid orbitals are directed towards the corners of a regular tetrahedron

CCB201

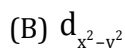
3. In which of the following process(s) hybridisation of underlined atom does not change ?



CCB223

Atomic Orbitals Involved in Hybridisation

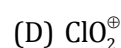
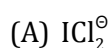
4. Which of the following 'd' orbital(s) participate in the hybridisation for Xe in the cationic part of XeF_6 (s) :-



CCB283

VSEPR Table

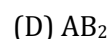
5. Which of the following species are linear ?



CCB226

6. Which of the following molecules have bent shape.

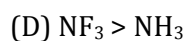
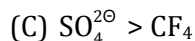
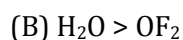
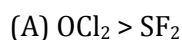
Where A is central atom, B is surrounding atom and E is lone pair -



CCB227

Bond Parameters; Bond Angle

7. INCORRECT order of bond angle is :



CCB224

Concept of Resonance; Bond Order; Comparison of Bond Length and Bond Strength using Bond Order

8. Select the correct statement -

- (A) C–O bond length of CO molecule is higher than the C–O bond length of CO₂
 (B) C–O bond length of CO molecule is lower than the C–O bond length of CO₂
 (C) N–O bond length of NO₃[⊖] ion is higher than the N–O bond length of NO₂[⊖]
 (D) N–O bond length of NO₃[⊖] ion is lower than the N–O bond length of NO₂[⊖]

CCB155

9. CO₃^{2⊖} anion has which of the following characteristics

- (A) Bonds of unequal length (B) sp² hybridization of C atom
 (C) Resonance stabilization (D) Same bond angles

CCB202

10. Select the CORRECT statement(s) :

- (A) Bond angle order : CH₄ > CF₄
 (B) Bond length order : d_{N–O}(NO₂[⊖]) < d_{N–O}(NO₃[⊖])
 (C) Bond order of S – O : SO₄^{2⊖} > SO₃^{2⊖}
 (D) Bond angle order : NH₃ < PH₃

CCB225

Drago's Rule

11. In which of the following, the theory of hybridisation does not help to predict the bond angle ?

- (A) PH₃ (B) SbH₃ (C) SiH₄ (D) H₂S

CCB282

Odd Electron Molecules

12. Choose the correct options for NO₂ :

- (A) It is odd electron specie (B) N–O bond order = 1.5
 (C) Paramagnetic specie (D) Isoelectronic with CO₂

CCB254

13. The number of specie(s) which are not perfectly planar.

- (A) $\dot{\text{C}}\text{H}_3$ (B) $\dot{\text{C}}\text{F}_3$ (C) CHF₃ (D) CH₂F₂

CCB255

Molecules which Do Not Exist

14. Which of the following molecules do not exist ?

- (A) Br₂O (B) SF₆ (C) NCl₅ (D) OF₄

CCB284

15. Which of the following molecule exist-

- (A) SF₆ (B) IH₃ (C) SH₂ (D) PCl₅

CCB286

16. Which of the following molecules does not exist and non-existence reason is same as that of SH₆

- (A) OF₄ (B) SH₄ (C) PBr₆[⊖] (D) XeH₄

CCB287

Polarity of Species

17. Which of the following compounds possesses zero dipole moment?
 (A) Water (B) Benzene
 (C) Carbon tetrachloride (D) Boron trifluoride

CCB288

Types of Hydrogen Bonding

18. In which of the following chemical species hydrogen bonding takes places -
 (A) $\text{Na}_2\text{HPO}_3(\text{s})$ (B) $\text{K}_2\text{HPO}_4(\text{s})$ (C) $\text{KHCO}_3(\text{s})$ (D) $\text{NaHCO}_3(\text{s})$

CCB317

Applications of Hydrogen Bonding; Viscosity ; Boiling Point; Volatility; Solubility in Water

19. Which of the following statements are correct ?
 (A) The crystal lattice of ice is formed by covalent as well as hydrogen bonds
 (B) The density of water increases when heated from 0°C to 4°C
 (C) Above 4°C the thermal agitation of water molecules increases. Therefore, intermolecular distance increases and water starts expanding
 (D) The density of water decreases from 0°C to a maximum at 4°C

CCB315

Inert Pair Effect

20. Which of following molecules acts as good oxidizing agent ?
 (A) NaBiO_3 (B) PbO_2 (C) TlCl_3 (D) PbCl_2
21. Which of the following is CORRECT order of stability ?
 (A) $\text{Pb}^{2+} > \text{Pb}^{4+}$ (B) $\text{Bi}^{3+} > \text{Bi}^{5+}$ (C) $\text{Tl}^+ < \text{Tl}^{3+}$ (D) $\text{Sn}^{2+} > \text{Sn}^{4+}$

CCB527

CCB528

Back Bonding

22. Which of the following molecule(s) is/are having $p\pi-p\pi$ back bonding ?
 (A) BF_3 (B) BCl_3 (C) $\text{B}_3\text{N}_3\text{H}_6$ (D) None of these
23. Which of the following statement is/are correct for CCl_3^- and CCl_2 ?
 (A) Back bonding in both cases from Cl to C-atom
 (B) Back bonding in both cases from C to Cl-atom
 (C) Back bonding in CCl_3^- is from C to Cl but reverse in CCl_2 .
 (D) Direction of back bonding is just opposite in two cases.

CCB529

CCB530

Bridge Bonding(3C-2e- , 3C-4e-), B_2H_6 , Al_2Cl_6 , Bond Behaviour of B_2H_6 ; Bridge Bonding in other Compounds; Reactions of Diborane

24. Which of following are correct.
 (A) B_2H_6 is non-planar.
 (B) B_2H_6 undergo symmetrical cleavage with PF_3 , CO and $(\text{C}_2\text{H}_5)_3\text{N}$
 (C) B_2H_6 undergo unsymmetrical cleavage with NH_3 , CH_3NH_2 , $(\text{CH}_3)_2\text{NH}$.
 (D) $\text{BeH}_2(\text{s})$, Ga_2Me_6 , Al_2Me_6 have two type of bonds (2C-2e bond as well as 3C-2e bond)

CCB531

25. Which of the following molecule has incomplete octet?
 (A) B_2H_6 (B) Al_2Cl_6
 (C) Be_2Cl_4 (D) Be_2H_4 (vapor)

CCB532

Oxyacids of Sulphur

26. The acid having two directly connected sulphur atoms not present :
 (A) $H_2S_2O_7$ (B) $H_2S_2O_8$ (C) $H_2S_2O_6$ (D) H_2SO_5

CCB533

27. Which among the following is/are peroxo acid(s) ?
 (A) $H_2S_2O_3$ (B) H_2SO_5
 (C) $H_2S_2O_7$ (D) $H_2S_2O_8$

CCB534

Hydrolysis

28. When PCl_3 undergoes in hydrolysis then X oxyacid of phosphorus is formed then Select the correct statement about X :
 (A) Two P – OH bond (B) One P – H bond
 (C) One P = O bond (D) Three P – OH bond

CCB535

Introduction of Weak Forces; Types of Weak Forces; Factors Affecting London Dispersion Forces; Comparison of Boiling Point of Species

29. Which of the following factors are responsible for origination of Van Der Waals forces ?
 (A) Instantaneous dipole-induced dipole interaction
 (B) Dipole-induced dipole interaction
 (C) Dipole-dipole interaction
 (D) Size of molecule
30. Which of the following are true?
 (A) Van Der Waals forces are responsible for the formation of molecular crystals
 (B) Branching lowers the boiling points of isomeric organic compounds due to decrease in Van Der Waals forces of attraction
 (C) In graphite, Van Der Waals forces act between the carbon layers
 (D) In diamond, Van Der Waals forces act between the carbon layers

CCB335

CCB336

31. London forces depends upon :
 (A) Molecular weight (B) Number of polarisable electrons
 (C) Molecular size (D) None of these

CCB337

Linear Combination of Atomic Orbitals

32. Which of the following overlapping results non bonding molecular orbital (if overlapping axis is x-axis) ?
 (A) s of A & p_y of B (B) p_z of A & p_z of B
 (C) p_y of A & p_z of B (D) d_{xy} of A & s of B

CCB359

Bonding in some homonuclear diatomic molecules (Molecular Orbital Diagram of H_2 , He_2 , Li_2 , O_2 , B_2 , C_2 , N_2 , F_2 , Ne_2)

33. Which of the following statements are true for these given species : N_2 , CO , $CN^\ominus$ and $NO^\oplus$.
 (A) All species are paramagnetic (B) The species are isoelectronic
 (C) All the species have dipole moment (D) All the species are linear
 CCB360
34. Which of the following have unpaired electron(s)
 (A) $O_2^\oplus$ (B) $O_2^\ominus$ (C) NO (D) $H_2^\oplus$
 CCB361
35. Which of the following is/are paramagnetic ?
 (A) B_2 (B) O_2 (C) N_2 (D) He_2
 CCB362
36. Which of the following is/are correct?
 (A) During $N_2^\oplus$ formation, one electron is removed from the bonding molecular orbitals
 (B) During $O_2^\oplus$ formation, one electron is removed from the antibonding molecular orbitals
 (C) During $O_2^\ominus$ formation, one electron is added to the bonding molecular orbitals
 (D) During $N_2^\ominus$ formation, one electron is added to the bonding molecular orbitals
 CCB363
37. Which of the following statement is/are correct
 (A) The peroxide ion has a bond order of 1 while the oxygen molecule has a bond order of 2
 (B) The peroxide ion has a weaker bond than the dioxygen molecule has.
 (C) The peroxide ion as well as the dioxygen molecules are paramagnetic
 (D) The bond length of the peroxide ion is greater than that of the dioxygen molecule
 CCB380

Applications of MOT(Molecular Behaviour) ; 1 Bond Order; 2 Bond Length; 3 Bond Strength; 4 Ionisation Energy; 5 Colour

38. Which of the following have identical bond order ?
 (A) $O_2^{2\oplus}$ (B) $NO^\oplus$ (C) $CN^\ominus$ (D) $CN^\oplus$
 CCB379

Properties Affected by Polarisation; Melting Point of Ionic Compounds

39. Choose the correct order for the given properties.
 (A) $NaF < MgF_2 < AlF_3$: covalent character order.
 (B) $NaF < MgF_2 < AlF_3$: melting point order
 (C) $NaF < MgF_2 < AlF_3$: lattice energy order
 (D) $NaF > MgF_2 > AlF_3$: order of polarising power of cation.
 CCB416

Properties Affected by Polarisation; Solubility of Ionic Compounds in Water

40. Choose the correct order(s) for the given properties.
 (A) $MgSO_4 < SrSO_4 < BaSO_4$: Thermal stability order
 (B) $BeC_2O_4 < CaC_2O_4 < BaC_2O_4$: Solubility in water
 (C) $LiCl > NaCl > KCl$: Melting point order
 (D) $BeF_2 > CaF_2 > SrF_2$: Covalent character order
 CCB414

ASSERTION & REASON TYPE QUESTIONS

Polarity of Species

41. Statement-1 : Dipole moment of H_2O is more than that of OF_2 .
Statement-2 : In H_2O , the resultant bond dipole moment of O – H bond and the resultant lone pair dipole moment are in opposite direction.
(A) Statement-1 is true, statement-2 is true and statement-2 is correct explanation for statement-1.
(B) Statement-1 is true, statement-2 is true and statement-2 is NOT the correct explanation for statement-1.
(C) Statement-1 is true, statement-2 is false.
(D) Statement-1 is false, statement-2 is true.

CCB303

Types of Hydrogen Bonding

42. Statement-1 : p-Hydroxybenzoic acid has a lower boiling point than o-hydroxybenzoic acid.
Statement-2 : o-Hydroxybenzoic acid has intramolecular hydrogen bonding.
(A) Statement-1 is True, Statement-2 is True; Statement-2 is a correct explanation for Statement-1.
(B) Statement-1 is True, Statement-2 is True; Statement-2 is NOT a correct explanation for Statement-1.
(C) Statement-1 is True, Statement-2 is False.
(D) Statement-1 is False, Statement-2 is True.

CCB302

Linear Combination of Atomic Orbitals

43. Statement-1 : The melting point of noble gases increases as its atomic mass increases.
Statement-2 : Instantaneous dipole induced dipole attraction increases with increase in atomic mass of noble gases.
(A) Statement-1 is true, statement-2 is true and statement-2 is correct explanation for statement-1.
(B) Statement-1 is true, statement-2 is true and statement-2 is NOT the correct explanation for statement-1.
(C) Statement-1 is true, statement-2 is false.
(D) Statement-1 is false, statement-2 is true.

CCB338

44. Statement-1 : CCl_4 has lower boiling point than that of SiCl_4 .
Statement-2 : The magnitude of negative charge developed at chlorine atoms in SiCl_4 is more in comparison to negative charge developed at chlorine atoms in CCl_4 .
(A) Statement-1 is true, statement-2 is true and statement-2 is correct explanation for statement-1.
(B) Statement-1 is true, statement-2 is true and statement-2 is NOT the correct explanation for statement-1.
(C) Statement-1 is true, statement-2 is false.
(D) Statement-1 is false, statement-2 is true.

CCB339

Bonding in some homonuclear diatomic molecules (Molecular Orbital Diagram of H_2 , He_2 , Li_2 , O_2 , B_2 , C_2 , N_2 , F_2 , Ne_2)

45. Statement-1 : Super oxide ion is paramagnetic whereas peroxide ion is diamagnetic.
Statement-2 : Super oxide ion has one unpaired electron whereas peroxide ion has no unpaired electron.
(A) Statement-1 is true, statement-2 is true and statement-2 is correct explanation for statement-1.
(B) Statement-1 is true, statement-2 is true and statement-2 is NOT the correct explanation for statement-1.
(C) Statement-1 is true, statement-2 is false.
(D) Statement-1 is false, statement-2 is true.

CCB346

46. Statement-1: H_2 molecule is more stable than He-H molecule.
Statement-2: The antibonding electron in He-H molecule decreases the bond order and there by the stability.
(A) Statement-1 is true, statement-2 is true and statement-2 is correct explanation for statement-1.
(B) Statement-1 is true, statement-2 is true and statement-2 is NOT the correct explanation for statement-1.
(C) Statement-1 is true, statement-2 is false.
(D) Statement-1 is false, statement-2 is true.

CCB347

Concept of Hydration; Hydration Energy; Hydrated Radius; Ionic Mobility in Water; Conductivity in Water; Condition for Solubility of Ionic Compounds in Water

47. **Statement-1** : Among alkali metal cations, $Li^+(aq.)$ has highest electrical conductance.
Statement-2 : $Li^+(aq.)$ is largest alkali metal cation because of greater degree of hydration.
(A) Statement-1 is true, statement-2 is true and statement-2 is correct explanation for statement-1.
(B) Statement-1 is true, statement-2 is true and statement-2 is NOT the correct explanation for statement-1
(C) Statement-1 is true, statement-2 is false.
(D) Statement-1 is false, statement-2 is true.

CCB394

COMPREHENSION TYPE QUESTIONS

Polarity of Species

Paragraph for Questions: 48 & 49

Bond angle and bond polarity are important parameters associated for bonding in molecule.

48. Which of the following bond angle order is incorrect.
(A) $NH_3 > PH_3$ (B) $BF_3 < BH_3$
(C) $O(CH_3)_2 < O(C_2H_5)_2$ (D) $CO_3^{2-} < CO_2$

CCB292

49. Which of the following molecule is polar and non-planar
(A) CH_2Cl_2 (B) PCl_5
(C) SO_2 (D) XeF_4

CCB293

Applications of MOT(Molecular Behaviour) ; 1 Bond Order; 2 Bond Length; 3 Bond Strength; 4 Ionisation Energy; 5 Colour

Paragraph for Questions 50 & 51

Molecular orbital theory is based on linear combination of atomic orbitals (LCAO). According to LCAO when respective atomic orbitals of the atoms interact, they undergoes constructive and destructive interference giving two types of molecular orbital i.e. bonding and antibonding molecular orbitals respectively.

50. Which of the following specie is paramagnetic ?

- (A) $\text{NO}^\ominus$ (B) $\text{O}_2^{2\ominus}$ (C) $\text{CN}^\ominus$ (D) CO

CCB382

51. Bond order of Be_2 is:

- (A) 1 (B) 2 (C) 3 (D) 0

CCB383

MATRIX MATCH TYPE QUESTION

VSEPR Table

52. Select the correct matching:

List I

A : XeF_4

B : XeF_6

C : XeO_3

D : XeOF_2

List II

1. Pyramidal

2. T-shape

3. Distorted octahedral

4. Square planar

	A	B	C	D
(A)	4	3	1	2
(B)	1	2	3	4
(C)	2	1	3	4
(D)	4	1	3	2

CCB213

Bond Parameters; Bond Angle

53. Match list I with list II and select the correct answer:

List I (species)

(P) $\text{NO}_2^\oplus$

(Q) NO_2

(R) $\text{NO}_2^\ominus$

(S) $\text{NO}_3^\ominus$

List II (O-N-O angle)

(1) 180°

(2) 134°

(3) 120°

(4) 115°

(5) 109°

	P	Q	R	S
(A)	5	4	3	2
(B)	5	2	4	3
(C)	1	2	4	3
(D)	1	4	3	2

CCB253

Odd Electron Molecules

54. Match list I with list II and select the correct answer :

List-I

(Molecule / Species)

(P) NO₂(Q) ClO₂(R) ClO₃(S) CH₃

Code :

P	Q	R	S
(A) 2	4	1	3
(B) 2	1	3	4
(C) 1	4	2	3
(D) 3	1	2	4

List-II

(Unpaired electron resides in)

(1) d-orbital

(2) sp² - orbital(3) sp³ - orbital

(4) p-orbital

CCB243

Polarity of Species

55. Match the column

Column I

(A) I(CN)₂[⊖](B) CO₃^{2⊖}(C) XeO₂F₂(D) SOF₄

(A) (A) Q,S (B) P,R (C) Q,S (D) P,R,S

(C) (A) P,R,S (B) P,R (C) Q,S (D) Q,S

Column II

(P) Having pπ-pπ bond & μ_D = 0(Q) Having pπ-dπ bond & μ_D ≠ 0

(R) Planar

(S) Central atom is sp³d

(B) (A) P,R,S (B) Q,S (C) P,R (D) Q,S

(D) (A) Q,S (B) P,R (C) P,R,S (D) Q,S

CCB281

56. Column-I

(1) 2 lone pair on central atom

(2) Zero dipole moment

(3) Planar

(4) All adjacent bond angles are equal

(A) (1) P, Q; (2) P, R, S; (3) P, R, S; (4) P, Q, R

(C) (1) R, S; (2) P, Q, S; (3) P, R, S; (4) P, Q, S

Column-II

(P) XeF₅[⊖](Q) NF₃(R) C₂H₄(S) XeF₄

(B) (1) P, S; (2) P, R, S; (3) P, R, S; (4) P, Q, S

(D) (1) R, S; (2) P, R, S; (3) P, R, S; (4) P, Q, R

CCB304

Bridge Bonding(3C-2e⁻, 3C-4e⁻), B₂H₆, Al₂Cl₆, Bond Behaviour of B₂H₆; Bridge Bonding in other Compounds; Reactions of Diborane

57. Match the column :

Column I

(A) B₂H₆(B) Be₂H₄(C) Be₂Cl₄(D) Al₂(CH₃)₆

(A) A → P, R; B → Q, S; C → P, S; D → Q, R

(C) A → Q, R; B → Q, S; C → P, S; D → P, R

Column II

(P) (3c-4e[⊖]) bond(Q) (3c-2e[⊖]) bond(R) Central atom is sp³ hybridised(S) central atom is sp² hybridised.

(B) A → Q, R; B → P, S; C → P, Q; D → P, R

(D) A → S, R; B → R, S; C → P, S; D → P, R

CCB536

Introduction of Weak Forces; Types of Weak Forces; Factors Affecting London Dispersion Forces; Comparison of Boiling Point of Species

58. Match list I with list II and select the correct answer:

List-I

(Process)

- (P) Clathrate compound of Xe in ice
(Q) Liquation of Xe gas
(R) Liquation of HCl gas
(S) Hydration of $\text{Na}^{\oplus}$

List-II

(Operating interaction involved)

- (1) Ion - Dipole
(2) Dipole - Dipole
(3) Dipole - Induced dipole
(4) London forces

Code :

P	Q	R	S
(A) 1	3	4	2
(B) 3	4	1	2
(C) 3	4	2	1
(D) 1	3	2	4

CCB328

TABLE TYPE QUESTION

Linear Combination of Atomic Orbitals

Direction for Q.59 to Q.60 by appropriately matching the information given in the three columns of the following table

Molecular Orbital	Number of nodal plane	Symmetry of Molecular Orbital
(1) σ_s	(P) 3	(I) BMO, gerade
(2) π_p	(Q) 2	(II) ABMO, ungerade
(3) σ_{pz}^*	(R) 1	(III) BMO, ungerade
(4) δ^*	(S) 0	(IV) ABMO, gerade

59. Which of the following matching is **INCORRECT**.

- (A) (1), S, I (B) (2), R, I (C) (3), R, II (D) (4), P, II

CCB384

60. How many orbitals are occupied with set (1)(S)(I) in F_2

- (A) 0 (B) 1 (C) 2 (D) 3

CCB385

EXERCISE (S)

Part-1 : JEE Main (Numerical Value type)

- The answer to each question is a **Numerical Value**.
- For each question, enter the correct integer value (In case of non-integer value, the answer should be **rounded off** to the nearest Integer).

Kossel-Lewis approach to Chemical Bonding; Lewis Dot Structure

1. Draw the Lewis dot structure of the given compounds and find how many compounds contain lone pair(s) on central atom ?

- | | | | |
|------------------------------|-------------------------------|--------------------------|---------------------------------|
| 1. $\text{CCl}_3^\ominus$ | 2. CCl_4 | 3. $\text{NO}_2^\ominus$ | 4. $\text{NO}_3^\ominus$ |
| 5. CO | 6. CO_2 | 7. $\text{SCN}^\ominus$ | 8. $\text{SnCl}_3^\ominus$ |
| 9. $\text{BH}_4^\ominus$ | 10. $\text{BeF}_4^{2\ominus}$ | 11. SO_3 | 12. $\text{H}_3\text{O}^\oplus$ |
| 13. $\text{CO}_3^{2\ominus}$ | | | |

CCB157

Limitations of Lewis Dot Structure (Hypervalent, Hypovalent, Odd Electron Species)

2. How many of the following compounds are exception to the octet rule ?

H_2SO_4 , BeCl_2 , ClO_2 , H_2O , H_2 , CCl_4 , HF , NH_3 , BF_3

CCB156

Types of Covalent Bond and Overlapping of Orbitals; Orbital Overlap along Different Axes

3. Which of the given set of orbitals form ' σ ' bond between two atoms along y-axis ?

$s + p_x$, $s + p_y$, $p_y + p_y$, $p_x + p_y$, $p_y + p_z$, $p_x + p_z$

CCB181

Prediction of Hybridisation; Types of Hybridisation

4. In which of the given species central atom is sp^3 hybridized ?

SF_4 , POCl_3 , H_2O , XeO_3 , PCl_5 , OF_2 , $\text{SO}_3^{2\ominus}$, XeO_3F_2 , XeF_4

CCB203

Calculation of Number of p pi-p pi and p pi-d pi Bonds

5. Find the number of $p\pi - p\pi$ bond in $\text{IO}_3^\ominus$:

CCB204

6. Find out total number of $p\pi - d\pi$ bonds present in SO_2 molecule :

CCB205

Solid State Hybridisation

7. What is the bond angle difference between anionic part of solid Iodine molecule and anionic part of $\text{N}_2\text{O}_5(\text{s})$

CCB256

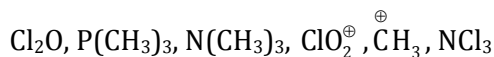
VSEPR Table

8. How many compounds have distorted shape ?

CH_4 , CHCl_3 , CH_2Cl_2 , SF_6 , BF_2Cl

CCB228

9. Among the following total number of planar molecules is / are :



CCB234

Maximum Atoms in A Plane; Number of Such Planes

10. Among the given species, how many species have maximum '3' atoms lying in a plane?
 $\text{BeCl}_2, \text{SnCl}_2, \text{SF}_2, \text{XeO}_3\text{F}_2, \text{IF}_7$

CCB229

11. Find the maximum number of atoms lying in a plane in AsF_5 :

CCB231

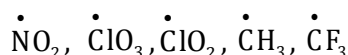
Bond Parameters; Bond length

12. Find the total number of following molecule(s) which have all bond lengths are same :
 $\text{XeF}_4, \text{SF}_4, \text{SH}_2, \text{NO}_3^\ominus, \text{SiF}_4, \text{ClF}_3, \text{PF}_2\text{Cl}_3, \text{XeO}_3\text{F}_2$

CCB232

Odd Electron Molecules

13. Find the number of species which have tendency to dimerize :



CCB257

Dipole Moment; Calculation of Percent Ionic Character

14. A diatomic molecule has a dipole moment of 1.2 D. If bond length is $1.0 \text{ \AA}$, what percentage of an electronic charge exists on each atom.
 [Add the digits untill you are getting single digit. e.g. If the answer is 537, then $5 + 3 + 7 = 15$] again $1 + 5 = 6$ and 6 is to be written as answer.

CCB294

Polarity of Species

15. Among the given molecules, number of molecules having some net polarity ($\mu \neq 0$)
 $\text{XeF}_4, \text{SO}_2, \text{SF}_4, \text{CH}_2\text{Cl}_2, \text{NF}_3, \text{BF}_3$

CCB417

Types of Hydrogen Bonding

16. Number of species from the following which shows hydrogen bonding.
 Chloral, HF, H_2O_2 , $\text{HPO}_3^{2\ominus}$, HNO_3 , H_3BO_3 , NaHCO_3 , $\text{HF}_2^\ominus$

CCB318

Back Bonding

17. In how many of the following species back bonding is possible ?
 $\text{PH}_3, \text{BF}_3, \text{BCl}_3, \text{PF}_3, \text{BF}_4^\ominus, \text{N}(\text{SH}_3)_3, \text{N}(\text{CH}_3)_3$

CCB537

Bridge Bonding(3C-2e-, 3C-4e-), B_2H_6 , Al_2Cl_6 , Bond Behaviour of B_2H_6 ; Bridge Bonding in other Compounds; Reactions of Diborane

18. Number of lone pair present in Al_2Cl_6 .

CCB538

19. Total number of Non-ionisable Hydrogen atom in H_3PO_3 .

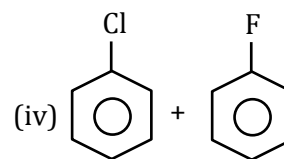
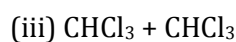
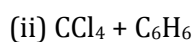
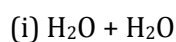
CCB539

20. Basicity of $\text{H}_4\text{P}_2\text{O}_7$.

CCB540

Introduction of Weak Forces; Types of Weak Forces; Factors Affecting London Dispersion Forces; Comparison of Boiling Point of Species

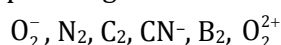
21. Dipole-Dipole interaction is present in how many of the given species ?



CCB419

Energy level diagram for s-p mixing and without s-p mixing case; Electronic Configuration

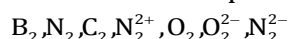
22. If s and p-orbital mixing is operative for all species given, then how many following species is/are paramagnetic in nature?



CCB420

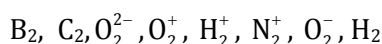
Bonding in some homonuclear diatomic molecules (Molecular Orbital Diagram of $\text{H}_2, \text{He}_2, \text{Li}_2, \text{O}_2, \text{B}_2, \text{C}_2, \text{N}_2, \text{F}_2, \text{Ne}_2$)

23. Total number of species in which highest occupied molecular orbital has gerade type symmetry.



CCB421

24. How many of the following molecule(s) or ion(s) is/are having bond order less than or equal to two?



CCB422

Applications of MOT (Molecular Behaviour) ; 1 Bond Order; 2 Bond Length; 3 Bond Strength; 4 Ionisation Energy; 5 Colour

25. Find the number of Chemical species which are paramagnetic in nature.



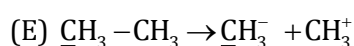
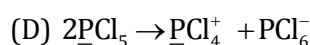
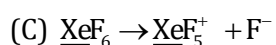
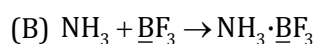
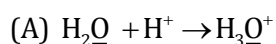
CCB424

Part-2 : JEE Advanced (Decimal type)

- The answer to each question is a **NUMERICAL VALUE**.
- For each question, enter the correct numerical value of the answer using the mouse and the onscreen virtual numeric keypad in the place designated to enter the answer.
- If the numerical value has more than two decimal places, **truncate/round-off** the value to **TWO** decimal places.

Prediction of Hybridisation; Types of Hybridisation

26. Find our number of transformation among following which involves the change of hybridisation of underlines atom.



CCB505

Maximum Atoms in A Plane; Number of Such Planes

27. Find the maximum number of plane having maximum number of atoms in CCl_4 .

CCB506

Concept of Resonance; Bond Order; Comparison of Bond Length and Bond Strength using Bond Order

28. If $x = \text{C-O bond order in } \text{CO}_3^{2-}$
 $y = \text{C-O bond order in } \text{HCOO}^-$
 $z = \text{Cl-O bond order in } \text{ClO}_4^-$
 Then $x + y + z = \dots$

CCB507

Dipole Moment; Calculation of Percent Ionic Character

29. Experimental dipole moment of HCl molecule is 1.03 D and bond distance is 1.275 Å. Calculate the value of $x/10$. Where x is % of ionic character :

CCB508

30. Calculate the percentage of covalent character of HX having bond length = 1.62 Å and observed dipole moment = 0.39 D.

CCB509

Back Bonding

31. Find the number molecules in which bond angle is changed due to back bonding w.r.t B/O/N.
 H_3BO_3 , $\text{B}(\text{OMe})_3$, BF_3 , $\text{N}(\text{SiH}_3)_3$, $\text{O}(\text{SiH}_3)_2$

CCB510

Oxyacids of Phosphorus

32. Find the value of z in $\text{H}_4\text{P}_2\text{O}_8$.

$$z = \frac{x + d}{a + c + b}$$
 Total number of σ bond = x
 Total number of lone pair of e-s are = a
 Total number of sp^3 hybridised atoms are b
 Total number of $\text{p}\pi\text{-p}\pi$ bonds are c
 Total number of $\text{p}\pi\text{-d}\pi$ bonds are d

CCB511

Oxyacids of Sulphur

33. Find the value of $\frac{\text{sigma bond} + \text{sum of total lone pair } e^-}{\text{Total number of } \pi \text{ bond}}$ for Marshall's acid.

CCB512

Applications of MOT(Molecular Behaviour) ; 1 Bond Order; 2 Bond Length; 3 Bond Strength; 4 Ionisation Energy; 5 Colour

34. There are some species given below.

(i)	O_2^+	(ii)	CO	(iii)	B_2	(iv)	O_2^-	(v)	NO^+
(vi)	He_2^+	(vii)	C_2^+	(viii)	CN^-	(ix)	N_2^-		

Total number of species which have their fractional bond order

CCB513

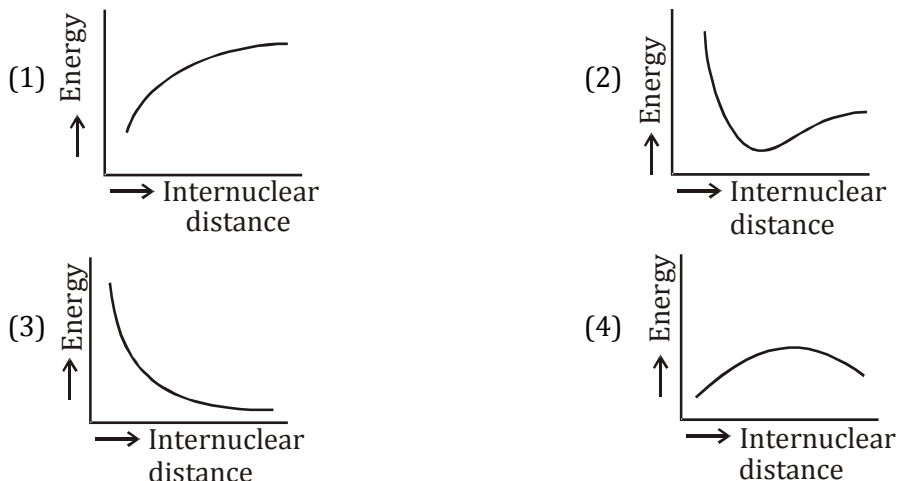
35. Total number of P—H bonds in cyclotrimetaphosphoric acid.

CCB541

EXERCISE - JEE (Main) PYQ

1. The potential energy curve for the H_2 molecule as a function of internuclear distance is :

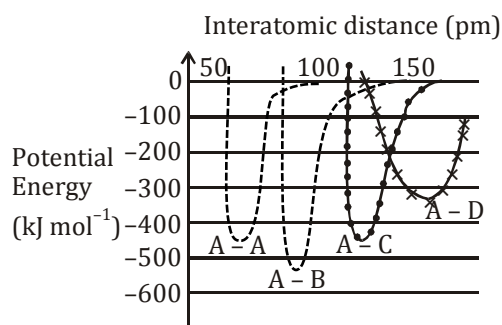
[JEE (Main) 2020]



CCB039

2. The intermolecular potential energy for the molecules A, B, C and D given below suggests that :

[JEE (Main) 2020]



- (1) D is more electronegative than other atoms
- (2) A-D has the shortest bond length
- (3) A-B has the stiffest bond
- (4) A-A has the largest bond enthalpy

CCB043

3. Match the LIST-I with LIST-II

LIST-I (Classification of molecules based on octet rule)		LIST-II (Example)	
A.	Molecules obeying octet rule	I.	NO, NO_2
B.	Molecules with incomplete octet	II.	$BCl_3, AlCl_3$
C.	Molecules with incomplete octet with odd electron	III.	H_2SO_4, PCl_5
D.	Molecules with expanded octet	IV.	CCl_4, CO_2

Choose the **correct** answer from the options given below :

[JEE (Main) 2025]

- (1) A-IV, B-II, C-I, D-III
- (2) A-III, B-II, C-I, D-IV
- (3) A-IV, B-I, C-III, D-II
- (4) A-II, B-IV, C-III, D-I

CCB598

4. Number of compounds with one lone pair of electrons on central atom amongst following is _
O₃, H₂O, SF₄, ClF₃, NH₃, BrF₅, XeF₄ [JEE (Main) 2024]
CCB514
5. The hybridisations of the atomic orbitals of nitrogen in NO₂[○], NO₂[⊕] and NH₄[⊕] respectively are.
(1) sp³, sp² and sp (2) sp, sp² and sp³ (3) sp³, sp and sp² (4) sp², sp and sp³ [JEE (Main) 2021]
CCB047
6. The structure of PCl₅ in the solid state is :
(1) Square pyramidal
(2) Tetrahedral [PCl₄][⊕] and octahedral [PCl₆][⊖]
(3) Square planar [PCl₄][⊕] and octahedral [PCl₆][⊖]
(4) Trigonal bipyramidal [JEE (Main) 2020]
CCB038
7. The number of molecules/ions that show linear geometry among the following is _____.
SO₂, BeCl₂, CO₂, N₃⁻, NO₂, F₂O, XeF₂, NO₂⁺, I₃⁻, O₃ [JEE (Main) 2025]
CCB599
8. The molecules having square pyramidal geometry are
(1) BrF₅ & XeOF₄ (2) SbF₅ & XeOF₄
(3) SbF₅ & PCl₅ (4) BrF₅ & PCl₅ [JEE (Main) 2025]
CCB600
9. Consider 'n' is the number of lone pair of electrons present in the equatorial position of the most stable structure of ClF₃. The ions from the following with 'n' number of unpaired electrons are :
A. V³⁺ B. Ti³⁺
C. Cu²⁺ D. Ni²⁺
E. Ti²⁺
Choose the **correct** answer from the options given below : [JEE (Main) 2025]
(1) A and C only (2) A, D and E only
(3) B and C only (4) B and D only CCB601
10. Which of the following statement is true with respect to H₂O, NH₃ and CH₄?
A. The central atoms of all the molecules are sp³ hybridized.
B. The H–O–H, H–N–H and H–C–H angles in the above molecules are 104.5°, 107.5° and 109.5° respectively.
C. The increasing order of dipole moment is
CH₄ < NH₃ < H₂O.
D. Both H₂O and NH₃ are Lewis acids and CH₄ is a Lewis base
E. A solution of NH₃ in H₂O is basic. In this solution NH₃ and H₂O act as Lowry-Bronsted acid and base respectively.
Choose the correct answer from the options given below : [JEE (Main) 2025]
(1) A, B and C only (2) C, D and E only
(3) A, D and E only (4) A, B, C and E only CCB602

11. Match List I and with List II

List-I (Molecule)		List-II (Shape)	
A	NH ₃	I	Square pyramid
B	BrF ₅	II	Tetrahedral
C	PCl ₅	III	Trigonal pyramidal
D	CH ₄	IV	Trigonal bipyramidal

Choose the **correct** answer from the option below:

[JEE (Main) 2024]

(1) A-IV, B-III, C-I, D-II (2) A-II, B-IV, C-I, D-III (3) A-III, B-I, C-IV, D-II (4) A-III, B-IV, C-I, D-II

CCB515

12. Match List I with List II

LIST-I (molecules/ions)		LIST-II (No. of lone pairs of e ⁻ on central atom)	
(A)	IF ₇	I	Three
(B)	ICl ₄ ⁻	II	One
(C)	XeF ₆	III	Two
(D)	XeF ₂	IV	Zero

Choose the **correct** answer from the options given below:

[JEE (Main) 2023]

(1) A - II, B - III, C - IV, D - I

(2) A - IV, B - III, C - II, D - I

(3) A - II, B - I, C - IV, D - III

(4) A - IV, B - I, C - II, D - III

CCB426

13. Amongst the following, the number of species having the linear shape is _____.

XeF₂, I₃⁺, C₃O₂, I₃⁻, CO₂, SO₂, BeCl and BCl₂[⊖]

[JEE (Main) 2023]

CCB427

14. Match List-I with List-II :

List-I Species	List-II Geometry/Shape
A. H ₃ O ⁺	I. Tetrahedral
B. Acetylide anion	II. Linear
C. NH ₄ ⁺	III. Pyramidal
D. ClO ₂ ⁻	IV. Bent

Choose the correct answer from the options given below :

[JEE (Main) 2023]

(1) A-III, B-II, C-I, D-IV (2) A-III, B-I, C-II, D-IV (3) A-III, B-IV, C-I, D-II (4) A-III, B-IV, C-II, D-I

CCB429

15. Match List - I with List - II.

List - I
(Compound)(A) BrF₅(B) [CrF₆]^{3⊖}(C) O₃(D) PCl₅List - II
(Shape)

(I) bent

(II) square pyramidal

(III) trigonal bipyramidal

(IV) octahedral

Choose the **correct** answer from the options given below :

[JEE (Main) 2022]

(1) (A) - (I), (B) - (II), (C) - (III), (D) - (IV)

(2) (A) - (IV), (B) - (III), (C) - (II), (D) - (I)

(3) (A) - (II), (B) - (IV), (C) - (I), (D) - (III)

(4) (A) - (III), (B) - (IV), (C) - (II), (D) - (I)

CCB048

16. The shape/structure of $[\text{XeF}_5]^\ominus$ and XeO_3F_2 , respectively, are : [JEE (Main) 2020]
 (1) Pentagonal planar and trigonal bipyramidal
 (2) Trigonal bipyramidal and pentagonal planar
 (3) Octahedral and square pyramidal
 (4) Trigonal bipyramidal and trigonal bipyramidal
CCB045
17. The compound that has the largest H-M-H bond angle (M = N, O, S, C), is : [JEE (Main) 2020]
 (1) H_2O (2) CH_4 (3) NH_3 (4) H_2S
CCB040
18. Given below are two statements :
Statement(I) : Experimentally determined oxygen-oxygen bond lengths in the O_3 are found to be same and the bond length is greater than that of a $\text{O}=\text{O}$ (double bond) but less than that of a single ($\text{O}-\text{O}$) bond.
Statement (II) : The strong lone pair-lone pair repulsion between oxygen atoms is solely responsible for the fact that the bond length in ozone is smaller than that of a double bond ($\text{O}=\text{O}$) but more than that of a single bond ($\text{O}-\text{O}$).
 In the light of the above statements, choose the **correct** answer from the options given below:
[JEE (Main) 2025]
 (1) **Statement I** is true but **Statement II** is false
 (2) Both **Statement I** and **Statement II** are true
 (3) Both **Statement I** and **Statement II** are false
 (4) **Statement I** is false but **Statement II** is true
CCB603
19. Arrange the following compounds in increasing order of their dipole moment :
 HBr , H_2S , NF_3 and CHCl_3 [JEE (Main) 2025]
 (1) $\text{NF}_3 < \text{HBr} < \text{H}_2\text{S} < \text{CHCl}_3$ (2) $\text{HBr} < \text{H}_2\text{S} < \text{NF}_3 < \text{CHCl}_3$
 (3) $\text{H}_2\text{S} < \text{HBr} < \text{NF}_3 < \text{CHCl}_3$ (4) $\text{CHCl}_3 < \text{NF}_3 < \text{HBr} < \text{H}_2\text{S}$
CCB604
20. The number of non-polar molecules from the following is _____
 HF , H_2O , SO_2 , H_2 , CO_2 , CH_4 , NH_3 , HCl , CHCl_3 , BF_3 [JEE (Main) 2024]
CCB516
21. If AB_4 molecule is a polar molecule, a possible geometry of AB_4 is : [JEE (Main) 2020]
 (1) Square pyramidal (2) Tetrahedral
 (3) Square planar (4) Rectangular planar
CCB044
22. If the boiling point of H_2O is 373 K, the boiling point of H_2S will be : [JEE (Main) 2020]
 (1) Greater than 300 K but less than 373 K (2) Less than 300 K
 (3) Equal to 373 K (4) More than 373 K
CCB042
23. Choose the correct stability order of group 13 elements in their +1 oxidation state.
[JEE (Main) 2022]
 (1) $\text{Al} < \text{Ga} < \text{In} < \text{Tl}$ (2) $\text{Tl} < \text{In} < \text{Ga} < \text{Al}$ (3) $\text{Al} < \text{Ga} < \text{Tl} < \text{In}$ (4) $\text{Al} < \text{Tl} < \text{Ga} < \text{In}$
CCB542

24. The Lewis acid character of boron tri halides follows the order : [JEE (Main) 2023]
- (1) $\text{BBr}_3 > \text{BI}_3 > \text{BCl}_3 > \text{BF}_3$ (2) $\text{BCl}_3 > \text{BF}_3 > \text{BBr}_3 > \text{BI}_3$
- (3) $\text{BF}_3 > \text{BCl}_3 > \text{BBr}_3 > \text{BI}_3$ (4) $\text{BI}_3 > \text{BBr}_3 > \text{BCl}_3 > \text{BF}_3$

CCB543

25. Identify the correct statement for B_2H_6 from those given below. [JEE (Main) 2022]
- (A) In B_2H_6 , all B-H bonds are equivalent.
 (B) In B_2H_6 there are four 3-centre-2-electron bonds.
 (C) B_2H_6 is a Lewis acid.
 (D) B_2H_6 can be synthesized from both BF_3 and NaBH_4 .
 (E) B_2H_6 is a planar molecule.
- Choose the most appropriate answer from the options given below:
- (1) (A) and (E) only (2) (B), (C) and (E) only
 (3) (C) and (D) only (4) (C) and (E) only

CCB544

26. The correct statement about B_2H_6 is: [JEE (Main) 2021]
- (1) Terminal B-H bonds have less p-character when compared to bridging bonds.
 (2) The two B-H-B bonds are not of same length
 (3) All B-H-B angles are of 120°
 (4) Its fragment, BH_3 , behaves as a Lewis base

CCB545

27. Sum of π -bonds present in peroxodisulphuric acid and pyrosulphuric acid is [JEE (Main) 2023]
28. Consider the following sulphure based oxoacids.
 H_2SO_3 , H_2SO_4 , $\text{H}_2\text{S}_2\text{O}_8$ and $\text{H}_2\text{S}_2\text{O}_7$.
 Amongst these oxoacids, the number of those with peroxo(O-O) bond is ____.

CCB050

[JEE (Main) 2022]

CCB546

29. Match the type of interaction in Column A with the distance dependence of their interaction energy in Column B : [JEE (Main) 2020]

Column-A	Column-B
(I) Ion - ion	(a) $\frac{1}{r}$
(II) Dipole - dipole	(b) $\frac{1}{r^2}$
(III) London dispersion	(c) $\frac{1}{r^3}$
	(d) $\frac{1}{r^6}$

Code:

- (1) (I)-(a), (II)-(b), (III)-(c) (2) (I)-(a), (II)-(c), (III)-(d)
 (3) (I)-(a), (II)-(b), (III)-(d) (4) (I)-(b), (II)-(d), (III)-(c)

CCB041

30. Given below are two statements : One is labelled as **Assertion A** and the other is labelled as **Reason R**
Assertion A : Zero orbital overlap is an out of phase overlap.

Reason : It results due to different orientation/ direction of approach of orbitals.

In the light of the above statements. Choose the **correct** answer from the options given below

[JEE (Main) 2022]

- (1) Both A and R are true and R is the correct explanation of A
 (2) Both A and R are true but R is NOT the correct explanation of A
 (3) A is true but R is false
 (4) A is false but R is true

CCB049

31. Which of the following linear combination of atomic orbitals will lead to formation of molecular orbitals in homonuclear diatomic molecules [internuclear axis in z-direction] ?

- A. $2p_z$ and $2p_x$
 B. $2s$ and $2p_x$
 C. $3d_{xy}$ and $3d_{x^2-y^2}$
 D. $2s$ and $2p_z$
 E. $2p_z$ and $3d_{x^2-y^2}$

[JEE (Main) 2025]

- (1) E Only
 (2) A and B Only
 (3) D Only
 (4) C and D Only

CCB605

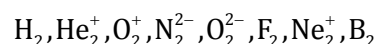
32. What is the number of unpaired electron(s) in the highest occupied molecular orbital of the following species : N_2 ; N_2^+ ; O_2 ; O_2^+ ?

[JEE (Main) 2023]

- (1) 0, 1, 2, 1
 (2) 2, 1, 2, 1
 (3) 0, 1, 0, 1
 (4) 2, 1, 0, 1

CCB425

33. The number of species from the following which are paramagnetic and with bond order equal to one is ____.



[JEE (Main) 2024]

CCB517

34. The bond order and magnetic property of acetylide ion are same as that of

[JEE (Main) 2023]

- (1) NO^+
 (2) O_2^+
 (3) O_2^-
 (4) N_2^+

CCB430

35. In which of the following processes, the bond order increases and paramagnetic character changes to diamagnetic one?

[JEE (Main) 2023]

- (1) $O_2 \rightarrow O_2^{2-}$
 (2) $NO \rightarrow NO^+$
 (3) $N_2 \rightarrow N_2^+$
 (4) $O_2 \rightarrow O_2^+$

CCB431

36. The number of following factors which affect the percent covalent character of the ionic bond is ____

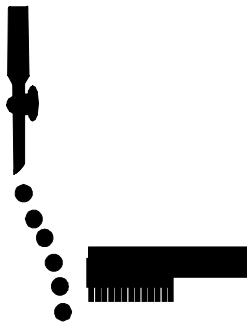
- (a) Polarising power of cation
 (b) Extent of distortion of anion
 (c) Polarisability of the anion
 (d) Polarising power of anion

[JEE (Main) 2023]

CCB428

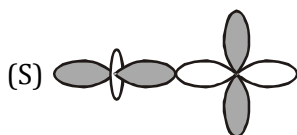
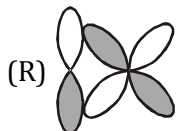
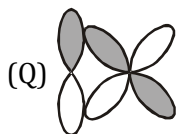
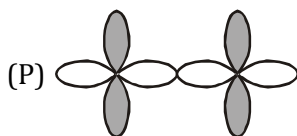
EXERCISE - JEE (Advanced) PYQ

1. Which ordering of compounds is according to the decreasing order of the oxidation state of Nitrogen ? [JEE (Advanced) 2012]
 (A) HNO_3 , NO , NH_4Cl , N_2 (B) HNO_3 , NO , N_2 , NH_4Cl
 (C) HNO_3 , NH_4Cl , NO , N_2 (D) NO , HNO_3 , NH_4Cl , N_2
- CCB075
2. The option(s) in which at least three molecules follow Octet Rule is(are) [JEE (Advanced) 2024]
 (A) CO_2 , C_2H_4 , NO and HCl (B) NO_2 , O_3 , HCl and H_2SO_4
 (C) BCl_3 , NO , NO_2 and H_2SO_4 (D) CO_2 , BCl_3 , O_3 and C_2H_4
- CCB518
3. The total number of lone pairs of electrons in N_2O_3 is : [JEE (Advanced) 2015]
CCB081
4. The total number of lone-pairs of electrons in Melamine is : [JEE (Advanced) 2013]
CCB077
5. In allene (C_3H_4), the type(s) of hybridisation of the carbon atoms is (are) : [JEE (Advanced) 2012]
 (A) sp and sp^3 (B) sp and sp^2
 (C) Only sp^2 (D) sp^2 and sp^3
- CCB074
6. Based on VSEPR model, match the xenon compounds given in List-I with the corresponding geometries and the number of lone pairs on xenon given in List-II and choose the correct option. [JEE (Advanced) 2024]
- | | List-I | | List-II |
|-----|--------------------------|-----|---|
| (P) | XeF_2 | (1) | Trigonal bipyramidal and two lone pair of electrons |
| (Q) | XeF_4 | (2) | Tetrahedral and one lone pair of electrons |
| (R) | XeO_3 | (3) | Octahedral and two lone pair of electrons |
| (S) | XeO_3F_2 | (4) | Trigonal bipyramidal and no lone pair of electrons |
| | | (5) | Trigonal bipyramidal and three lone pair of electrons |
- (A) $P \rightarrow 5, Q \rightarrow 2, R \rightarrow 3, S \rightarrow 1$ (B) $P \rightarrow 5, Q \rightarrow 3, R \rightarrow 2, S \rightarrow 4$
 (C) $P \rightarrow 4, Q \rightarrow 3, R \rightarrow 2, S \rightarrow 1$ (D) $P \rightarrow 4, Q \rightarrow 2, R \rightarrow 5, S \rightarrow 3$
- CCB519
7. The sum of the number of lone pairs of electrons on each central atom in the following species is : [JEE (Advanced) 2017]
 $[\text{TeBr}_6]^{2-}$, $[\text{BrF}_2]^+$, SNF_3 and $[\text{XeF}_3]^-$
 [Atomic number : $\text{N} = 7, \text{F} = 9, \text{S} = 16, \text{Br} = 35, \text{Te} = 52, \text{Xe} = 54$]
CCB086
8. The compound(s) with TWO lone pairs of electrons on the central atom is(are) : [JEE (Advanced) 2016]
 (A) BrF_5 (B) ClF_3
 (C) XeF_4 (D) SF_4
- CCB083

9. Among the triatomic molecules/ions, BeCl_2 , $\text{N}_3^\ominus$, N_2O , $\text{NO}_2^\oplus$, O_3 , SCl_2 , $\text{ICl}_2^\ominus$, $\text{I}_3^\ominus$ and XeF_2 the total number of linear molecules(s) / ion(s) where the hybridization of the central atoms does not have contribution from the d-orbital(s) is :
(Atomic number : S = 16, Cl = 17, I = 53 and Xe = 54) [JEE (Advanced) 2015]
CCB080
10. The shape of XeO_2F_2 molecule is : [JEE (Advanced) 2012]
(A) Trigonal bipyramidal (B) Square planar
(C) Tetrahedral (D) See-saw CCB076
11. Based on VSEPR theory, the number of 90 degree F–Br–F angles in BrF_5 is : [JEE (Advanced) 2010]
CCB072
12. Consider the following compounds in the liquid form :
 O_2 , HF, H_2O , NH_3 , H_2O_2 , CCl_4 , CHCl_3 , C_6H_6 , $\text{C}_6\text{H}_5\text{Cl}$
When a charged comb is brought near their flowing stream, how many of them show deflection as per the following figure ?
- 
- [JEE (Advanced) 2020]
CCB089
13. Each of the following options contains a set of four molecules. Identify the option(s) where all four molecules possess permanent dipole moment at room temperature :
(A) BeCl_2 , CO_2 , BCl_3 , CHCl_3 (B) SO_2 , $\text{C}_6\text{H}_5\text{Cl}$, H_2Se , BrF_5
(C) BF_3 , O_3 , SF_6 , XeF_6 (D) NO_2 , NH_3 , POCl_3 , CH_3Cl [JEE (Advanced) 2019]
CCB087
14. Among the following, the correct statement(s) is/are :- [JEE (Advanced) 2017]
(A) $\text{Al}(\text{CH}_3)_3$ has the three-centre two-electron bonds in its dimeric structure
(B) AlCl_3 has the three-centre two-electron bonds in its dimeric structure
(C) BH_3 has the three-centre two-electron bonds in its dimeric structure
(D) The Lewis acidity of BCl_3 is greater than that of AlCl_3 CCB547
15. The compound (s) having peroxide linkage is(are) :
(A) $\text{H}_2\text{S}_2\text{O}_7$ (B) $\text{H}_2\text{S}_2\text{O}_8$
(C) $\text{H}_2\text{S}_2\text{O}_5$ (D) H_2SO_5 [JEE (Advanced) 2024]
CCB520

16. Consider the following molecules : Br_3O_8 , F_2O , $\text{H}_2\text{S}_4\text{O}_6$, $\text{H}_2\text{S}_5\text{O}_6$, and C_3O_2 .
Count the number of atoms existing in their zero oxidation state in each molecule. Their sum is ____.
[JEE (Advanced) 2023]
CCB548
17. Among B_2H_6 , $\text{B}_3\text{N}_3\text{H}_6$, N_2O , N_2O_4 , $\text{H}_2\text{S}_2\text{O}_3$ and $\text{H}_2\text{S}_2\text{O}_8$, the total number of molecules containing covalent bond between two atoms of the same kind is :
[JEE (Advanced) 2019]
CCB088
18. The correct statement(s) regarding, (i) HClO , (ii) HClO_2 , (iii) HClO_3 and (iv) HClO_4 , is(are) :
[JEE (Advanced) 2015]
CCB082
- (A) The number of $\text{Cl} = \text{O}$ bonds in (ii) and (iii) together is two
(B) The number of lone pairs of electrons on Cl in (ii) and (iii) together is three
(C) The hybridization of Cl in (iv) is sp^3
(D) Amongst (i) to (iv), the strongest acid is (i)
19. For diatomic molecules, the correct statement(s) about the molecular orbitals formed by the overlap of two $2p_z$ orbitals is(are)
[JEE (Advanced) 2022]
CCB090
- (A) σ orbital has a total of two nodal planes.
(B) σ^* orbital has one node in the xz -plane containing the molecular axis.
(C) π orbital has one node in the plane which is perpendicular to the molecular axis and goes through the center of the molecule.
(D) π^* orbital has one node in the xy -plane containing the molecular axis.
20. Match the orbital overlap figures shown in List-I with the description given in List-II and select the correct answer using the code given below the lists :
[JEE (Advanced) 2014]

List-I



- (A) P-2, Q-1, R-3, S-4
(C) P-2, Q-3, R-1, S-4

List-II

(1) $p - d \pi$ antibonding

(2) $d - d \sigma$ bonding

(3) $p - d \pi$ bonding

(4) $d - d \sigma$ antibonding

- (B) P-4, Q-3, R-1, S-2
(D) P-4, Q-1, R-3, S-2

CCB079

21. Assuming 2s-2p mixing is NOT operative, the paramagnetic species among the following is :

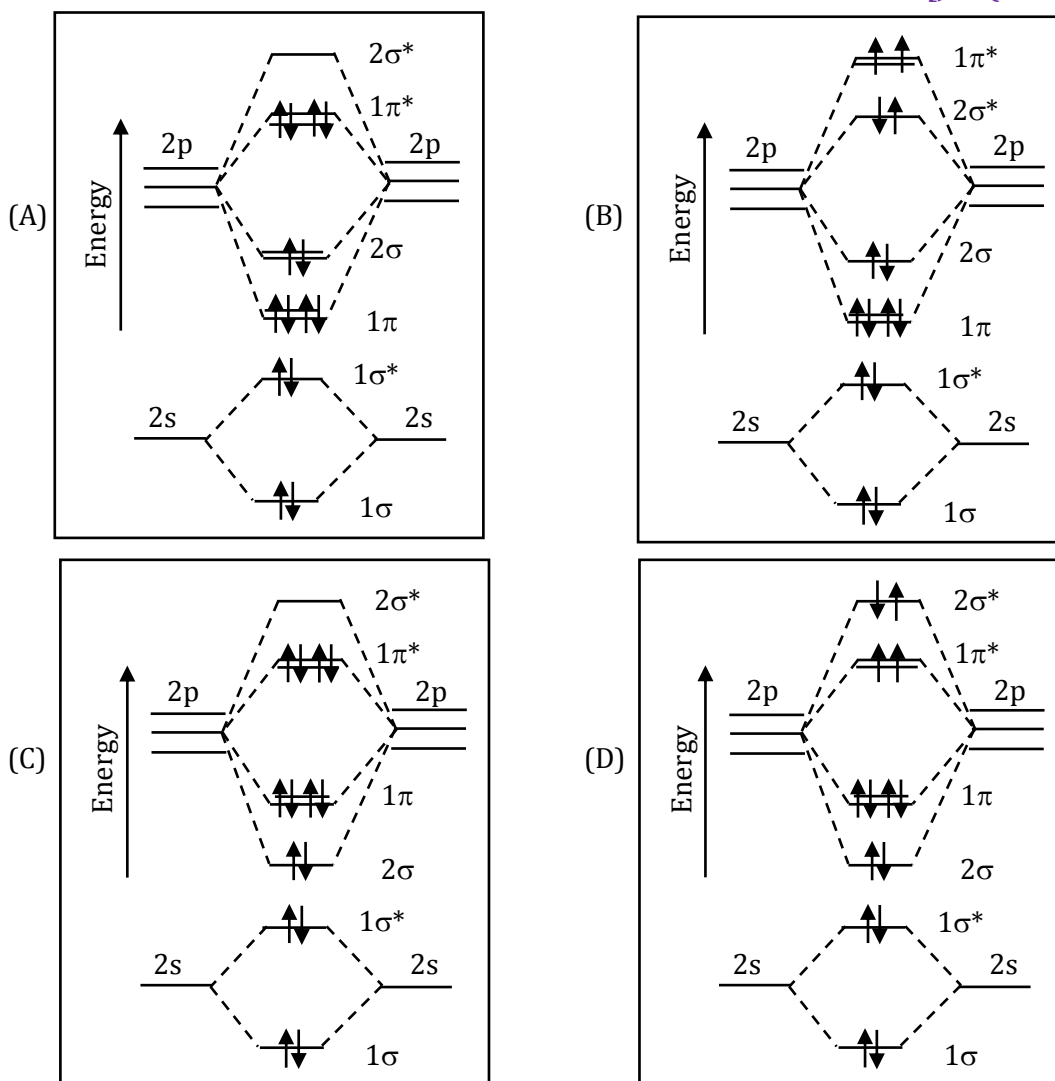
[JEE (Advanced) 2014]

- (A) Be_2 (B) B_2 (C) C_2 (D) N_2

CCB078

22. The correct molecular orbital diagram for F_2 molecule in the ground state is

[JEE (Advanced) 2023]



CCB432

23. Among H_2 , He_2^+ , Li_2 , Be_2 , B_2 , C_2 , N_2 , O_2^+ , and F_2 , the number of diamagnetic species is :
(Atomic number : $\text{H} = 1$, $\text{He} = 2$, $\text{Li} = 3$, $\text{Be} = 4$, $\text{B} = 5$, $\text{C} = 6$, $\text{N} = 7$, $\text{O} = 8$, $\text{F} = 9$)

[JEE (Advanced) 2017]

CCB085

24. According to Molecular Orbital Theory :

[JEE (Advanced) 2016]

- (A) C_2^{2+} is expected to be diamagnetic
(B) O_2^{2+} is expected to have a longer bond length than O_2
(C) N_2^+ and N_2^0 have the same bond order
(D) He_2^+ has the same energy as two isolated He atoms

CCB084

ANSWER KEY

EXERCISE (O-1)

Que.	1	2	3	4	5	6	7	8	9	10
Ans.	A	B	A	B	B	C	B	C	D	D
Que.	11	12	13	14	15	16	17	18	19	20
Ans.	D	A	C	A	B	A	D	A	D	D
Que.	21	22	23	24	25	26	27	28	29	30
Ans.	A	B	A	B	C	A	A	C	C	B
Que.	31	32	33	34	35	36	37	38	39	40
Ans.	B	C	C	A	B	C	A	B	B	B
Que.	41	42	43	44	45	46	47	48	49	50
Ans.	C	A	C	A	C	A	A	C	C	B
Que.	51	52	53	54	55	56	57	58	59	60
Ans.	A	A	C	C	C	D	B	B	B	C
Que.	61	62	63	64	65	66	67	68	69	70
Ans.	C	A	D	C	B	B	B	C	A	B
Que.	71	72	73	74	75	76	77	78	79	80
Ans.	C	B	C	C	B	B	A	D	C	A
Que.	81	82	83	84	85	86	87	88	89	90
Ans.	C	C	B	A	C	C	B	C	A	D
Que.	91	92	93	94	95	96	97	98	99	100
Ans.	A	A	D	C	B	D	C	B	A	A
Que.	101	102	103	104	105	106	107	108	109	110
Ans.	B	D	C	B	B	C	A	B	D	C
Que.	111	112	113	114	115	116	117	118	119	120
Ans.	A	C	C	B	C	D	D	C	B	B
Que.	121	122	123	124	125	126	127	128	129	130
Ans.	B	C	B	B	A	D	D	A	A	A
Que.	131	132	133	134	135	136	137	138	139	140
Ans.	C	B	A	B	B	D	B	B	C	B
Que.	141	142	143	144	145	146	147	148	149	150
Ans.	C	A	A	C	B	C	A	D	A	A
Que.	151	152	153	154	155	156	157	158	159	160
Ans.	A	D	D	A	B	C	A	B	B	C
Que.	161	162	163	164	165	166	167	168	169	170
Ans.	B	C	D	C	C	C	A	B	A	A
Que.	171	172	173	174	175	176	177	178	179	180
Ans.	A	C	A	D	D	A	B	B	C	A
Que.	181	182	183	184	185	186	187	188	189	190
Ans.	C	D	D	D	B	B	A	B	A	A
Que.	191	192	193	194	195	196	197	198	199	200
Ans.	C	B	C	B	B	B	D	D	D	A

EXERCISE (O-2)

Que.	1	2	3	4	5	6	7	8	9	10
Ans.	AC	AC	AC	BC	ABC	AB	CD	BC	BCD	BC
Que.	11	12	13	14	15	16	17	18	19	20
Ans.	ABD	ABC	BCD	CD	ACD	BD	BCD	BCD	ABC	ABC
Que.	21	22	23	24	25	26	27	28	29	30
Ans.	AB	ABC	CD	ABCD	ACD	ABD	BD	ABC	ABC	ABC
Que.	31	32	33	34	35	36	37	38	39	40
Ans.	ABC	ACD	BD	ABCD	AB	AB	ABD	ABC	ABC	AD
Que.	41	42	43	44	45	46	47	48	49	50
Ans.	C	D	A	D	A	A	D	B	A	A
Que.	51	52	53	54	55	56	57	58	59	60
Ans.	D	A	C	B	C	B	C	C	B	C

EXERCISE (S)

1.	5	2.	4	3.	2	4.	5	5.	0
6.	1	7.	60	8.	3	9.	3	10.	3
11.	4	12.	4	13.	4	14.	7	15.	4
16.	6	17.	4	18.	16	19.	1	20.	4
21.	3	22.	2	23.	4	24.	6	25.	4
26.	3.00	27.	10.00	28.	4.58	29.	1.68-1.70		
30.	95.00	31.	4.00	32.	0.62	33.	6.75		
34.	4.00	35.	0.00						

EXERCISE - JEE (Main) PYQ

Que.	1	2	3	4	5	6	7	8	9	10
Ans.	2	3	1	4	4	2	6	1	2	1
Que.	11	12	13	14	15	16	17	18	19	20
Ans.	3	2	5	1	3	1	2	1	1	4
Que.	21	22	23	24	25	26	27	28	29	30
Ans.	1	2	1	4	3	1	8	1	2	1
Que.	31	32	33	34	35	36				
Ans.	3	1	1	1	2	3				

EXERCISE - JEE (Advanced) PYQ

Que.	1	2	3	4	5	6	7	8	9	10
Ans.	B	AD	8	6	B	B	6	BC	4	D
Que.	11	12	13	14	15	16	17	18	19	20
Ans.	0	6	BD	ACD	BD	6	4	BC	AD	C
Que.	21	22	23	24						
Ans.	C	C	5 or 6	AC						